Power of a Point

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This article is about the various uses of Power of a Point within olympiad geometry. Sections 1 through 4 should be fairly accessible to beginners who have had some practice with angle chasing, although Section 5 requires some extra background on barycentric coordinates. If you don't know what something means, consult the first two Appendices or message me on AoPS; my username is Kagebaka.

Acknowledgements

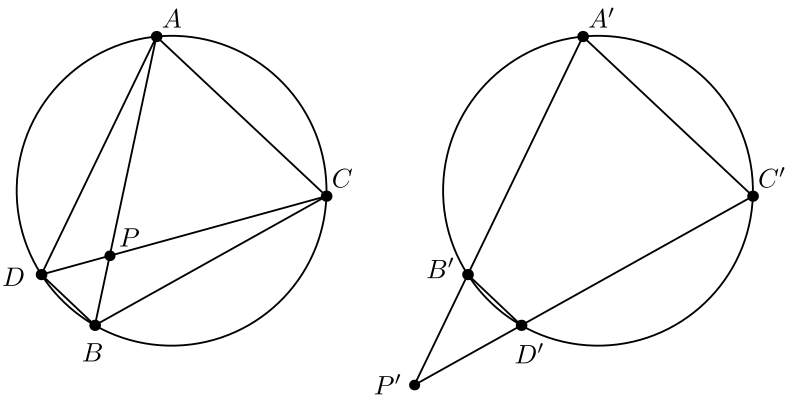
Special thanks to Evan Chen for providing the style file for this article, as well as AoPS user amar_04 for giving many problem suggestions and proofreading. Much of the material of this handout was inspired by Chapter 2 of Evan Chen's book *Euclidean Geometry in Mathematical Olympiads*, as well as Kenneth Peng's linearity handout.

Contents

§1 Introduction

First, we begin with the basic theorem, which allows us to derive cyclic quadrilaterals based on lengths:

Theorem 1.1 (Power of a Point)
Let A, B, C, D be points in the plane with $P = AB \cap CD$ such that P either lies on both segments AB, CD or neither. Then A, B, C, D are concyclic if and only if $PA \cdot PB = PC \cdot PD$.



Proof. Observe that

$$\begin{aligned} & PA \cdot PB = PC \cdot PD \\ \iff\ & \triangle PAC \overset{-}{\sim} \triangle PDB \\ \iff\ & \angle PAC = \angle PDB \\ \iff\ & A, B, C, D \text{ concyclic,} \end{aligned}$$

so we're done. □

Exercise 1.2. Prove the tangent case; i.e. if P, A, B are on a line in that order, then if C is one of the points on a circle passing through A, B such that PC is tangent to that circle, we must have $PA \cdot PB = PC^2$.

One observation we can make about this configuration is that the quantity $PA \cdot PB$ did not depend on the line AB, and just the position of point P with respect to the circle. This motivates the following definition:

Definition 1.3. The **power** of point P with respect to a circle ω with center O and radius R is

$$\text{Pow}(P, \omega) = OP^2 - R^2.$$

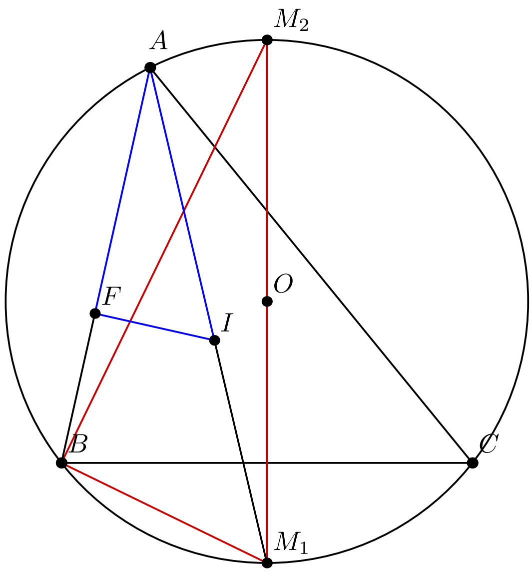
Note that this quantity actually has a sign (positive if P is outside ω, 0 if P is on ω, negative if P is inside ω), which is an important fact that you must remember!

Exercise 1.4. Check that the absolute value of the power of a point P returns the same product of lengths from the first theorem and the tangent case.

Actually, before we develop any further theory, we can already now solve some problems with just the basic PoP theorem:

Example 1.5 (Euler's Theorem)

Let $\triangle ABC$ be a triangle. If O and I are its circumcenter and incenter, respectively, and R and r are its circumradius and inradius, respectively, then prove that $OI^2 = R(R-2r)$, and in particular $R \geq 2r$.



Proof. It suffices to show that $R^2 - OI^2 = 2Rr$, which is clearly the power of point I with respect to the circumcircle of $\triangle ABC$. In particular, if we let AI meet the circumcircle again at M_1, the midpoint of the arc AB not containing A, then we just need to show that $AI \cdot IM_1 = 2Rr$. This looks sort of like similar triangles (remember this idea: similar triangles often come in handy when using PoP), so we are motivated to introduce the F, the point where the incircle touches AB, and the midpoint M_2 of arc BAC. Since $\angle BM_2M_1 = \angle BAM_1$ and $\angle M_2BM_1 = 90^\circ = \angle AFI$, we must have $\triangle AFI \sim \triangle M_2BM_1$, which gives us $\frac{AI}{2R} = \frac{BM_1}{r}$, but we know that $BM_1 = M_1I$ by the Incenter-Excenter Lemma, so we get the desired result upon cross multiplying. Of course, $R \geq 2r$ just follows from the fact that OI^2 and R are both positive, so $R - 2r$ must be as well unless $AB = BC = CA \implies O = I$, which gives the equality case. $\square$

Exercise 1.6. Let H be the orthocenter of $\triangle ABC$. Prove that

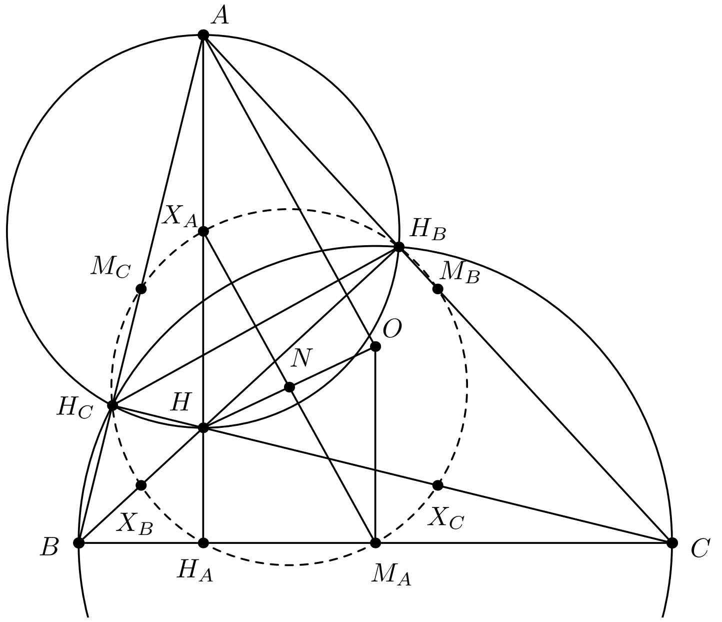
$$\text{Pow}(H, (ABC)) = 8R^2 \cos A \cos B \cos C.$$

Example 1.7 (Nine-Point Circle)

In a triangle $\triangle ABC$, let

- H be the orthocenter,
- H_A, H_B, H_C be the feet of the altitudes from A, B, C respectively,
- M_A, M_B, M_C be the midpoints of BC, CA, AC, respectively, and
- X_A, X_B, X_C be the midpoints of AH, BH, CH, respectively.

Then $H_A, H_B, H_C, M_A, M_B, M_C, X_A, X_B, X_C$ lie on a circle whose center is the midpoint of H and the circumcenter O (and in particular lies on the Euler Line).



Proof. The canonical proof of this theorem is with homothety, but we can still use Power of a Point in a very nice way to deal with most of the points. WLOG let $\triangle ABC$ be acute with $\angle B > \angle C$, and let H'_A, H'_B, H'_C be the reflections of H over BC, CA, AB, respectively, which are well-known to lie on (ABC). Then we have

$$\begin{aligned} -\mathrm{Pow}(H,(ABC)) = AH \cdot HH'_A = BH \cdot HH'_B = CH \cdot HH'_C \\ \implies \frac{1}{2}AH \cdot \frac{1}{2}HH'_A = \frac{1}{2}BH \cdot \frac{1}{2}HH'_B = \frac{1}{2}CH \cdot \frac{1}{2}HH'_C \\ \implies X_AH \cdot HH_A = X_BH \cdot HH_B = X_CH \cdot HH_C, \end{aligned}$$

so the converse of PoP implies that $H_A, H_B, H_C, X_A, X_B, X_C$ all lie on a circle ω.

Now, to prove that the midpoints also lie on ω, we observe that since X_A is the center of (AH_BHH_C), it lies on the perpendicular bisector of H_BH_C and we have $\angle H_CX_AH_B = 2\angle A$. On the other hand, we also know that M_A is the center of (BH_CH_BC), so M_A must also lie on the perpendicular bisector of H_BH_C and we have $\angle H_CM_AH_A = 2\angle H_CCH_B = 180^\circ - 2\angle A$. Thus, M_A lies on ω, and furthermore X_AM_A is a diameter.

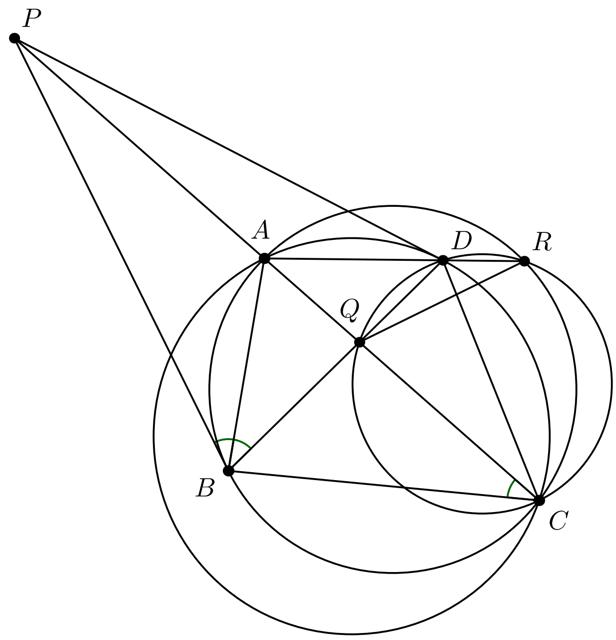
Finally, we want to show that the center N of ω is the midpoint of OH, which we can prove with some angle chasing. Observe that

$$\begin{aligned} \angle H_AX_AM_A &= 90^\circ - \angle X_AM_AH_A \\ &= 90^\circ - \angle X_AH_BH_A \\ &= 90^\circ - \angle X_AH_BH_C - \angle H_CH_BH_A \\ &= 90^\circ - \frac{1}{2}(180^\circ - 2\angle A) - 2(90^\circ - \angle B) \\ &= \angle A - \angle BAH_A - \angle OAC \\ &= \angle HAO, \end{aligned}$$

so $AO \parallel X_AM_A$ and in fact AX_AM_AO is a parallelogram as we also have $AX_A \perp BC \perp OM_A$. Since the opposite sides of a parallelogram are equal, we then must have $HX_A = AX_A = OM_A$, so $\triangle HX_AN \cong \triangle OM_AN$ by SAS congruence, and N is the midpoint of OH as desired. $\square$

Example 1.8 (IMO Shortlist 2013/G4)

Let ABC be a triangle with $\angle B > \angle C$. Let P and Q be two different points on line AC such that $\angle PBA = \angle QBA = \angle ACB$ and A is located between P and C. Suppose that there exists an interior point D of segment BQ for which $PD = PB$. Let the ray AD intersect the circle ABC at $R \neq A$. Prove that $QB = QR$.



Proof. The first thing we notice is that the angle conditions produce a number of tangencies: specifically, we have PB tangent to (ABC) and AB tangent to both (BQC) and (BDR). Now we would like to translate these tangencies into usable information, and PoP is the key: the former tangency implies that $PA \cdot PC = PB^2 = PD^2$ and the latter pair of tangencies implies that $AD \cdot AR = AB^2 = AQ \cdot AC$, so in fact PD is tangent to (ADC) and additionally $DQCR$ is cyclic, both by the converse of PoP.

Now the rest is just length chasing. By the Angle Bisector Theorem on $\triangle PBQ$, $QB = \frac{BP \cdot AQ}{AP}$, but PoP on A with respect to $(DQCR)$ and P with respect to (ABC) respectively tell us that $AQ = \frac{AD \cdot AR}{AC}$ and $AP = \frac{BP^2}{CP}$, so substituting gives us $QB = \frac{AR}{AC} \cdot AD \cdot \frac{CP}{BP}$. However, we also know that $\triangle ADC \sim \triangle AQR \implies \frac{QR}{CD} = \frac{AR}{AC}$ and $\triangle ADP \sim \triangle CDP \implies \frac{AD}{CD} = \frac{BP}{CP}$, so simplifying gives us

$$QB = \frac{QR}{CD} \cdot AD \cdot \frac{CP}{BP} = QR$$

as desired. □

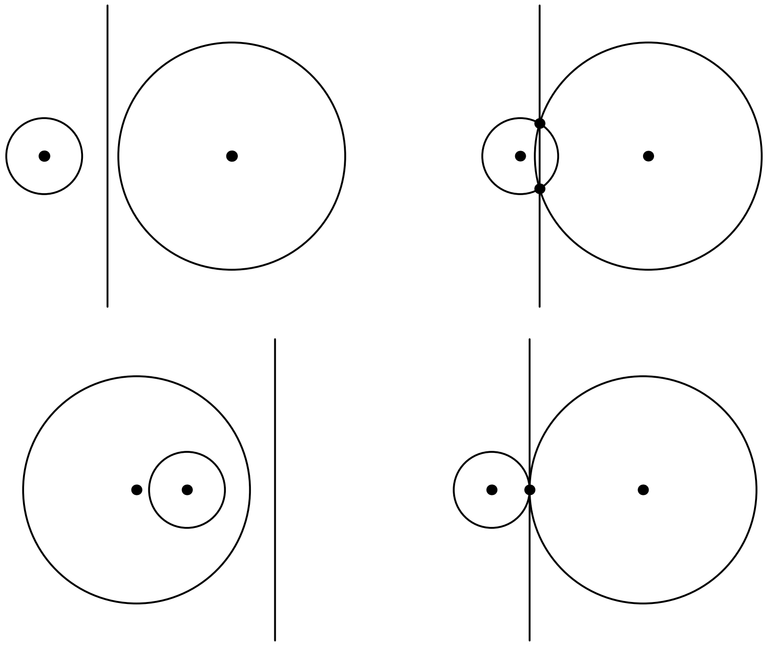
§2 Radical Axis

Now we look at what happens when the power of a point is equal with respect to two circles.

Definition 2.1. If ω_1 and ω_2 are two circles, then the set of points P such that $\text{Pow}(P, \omega_1) = \text{Pow}(P, \omega_2)$ is the **radical axis** of the two circles.

The reason why this locus is special is because of the following characterization:

Theorem 2.2 (Radical Axis)
The radical axis of two circles ω_1 and ω_2 with centers O_1 and O_2, respectively, is a line perpendicular to O_1O_2. In particular, the radical axis is the common chord if the two circles intersect and the common tangent if the two circles are tangent.



Proof. We use Cartesian coordinates. Let r_1 and r_2 be the radii of ω_1 and ω_2, respectively, and suppose that $O_1 = (a, 0), O_2 = (b, 0)$. Then for any point $P = (x, y)$, we have

$$\begin{aligned}\text{Pow}(P, \omega_1) &= (x-a)^2 + y^2 - r_1^2 \\ \text{Pow}(P, \omega_2) &= (x-b)^2 + y^2 - r_2^2,\end{aligned}$$

so

$$\begin{aligned}0 &= \text{Pow}(P, \omega_1) - \text{Pow}(P, \omega_2) \\ &= ((x-a)^2 + y^2 - r_1^2) - ((x-b)^2 + y^2 - r_2^2) \\ &= (-2a + 2b)x + (a^2 - b^2 + r_2^2 - r_1^2),\end{aligned}$$

which is the equation of a line perpendicular to the x-axis as desired. Also, if we let A, B are the intersections of ω_1 and ω_2, then $\text{Pow}(A, \omega_1) = 0 = \text{Pow}(B, \omega_2)$ and $\text{Pow}(A, \omega_1) = 0 = \text{Pow}(B, \omega_2)$, so AB must be the radical axis, and if ω_1 and ω_2 are tangent, then clearly their common tangent is perpendicular to O_1O_2 and their touch point has an equal power of 0 to both circles. $\square$

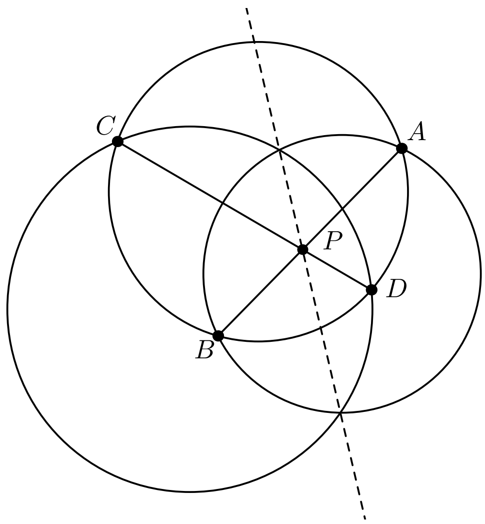
Exercise 2.3. Prove this theorem synthetically with the lemma that AB is perpendicular to CD if and only if $AC^2 + BD^2 = AD^2 + BC^2$.

> **Remark 2.4.** Observe that if we set r_1 or r_2 to 0, none of the algebra breaks. We will discuss the use of radical axes with degenerate circles in the fourth section.

> **Remark 2.5.** Furthermore, if we substitute the 0 with y, then $\text{Pow}(P, \omega_1) - \text{Pow}(P, \omega_2)$ is still a linear function in P. We will revisit this idea in the last section.

A natural extension of the radical axis is the **radical center**, which is the scenario where the power of a point is equal with respect to three circles rather than two. As we will see, this allows us to combine collinearity/concurrency with concyclicity.

Theorem 2.6 (Radical Center)
Let points A, B lie on ω_1 and points C, D lie on ω_2. Then $AB \cap CD = P$ lies on the radical axis of ω_1 and ω_2 if and only if $ABCD$ is a cyclic quadrilateral whose circumcenter does not lie on the line connecting the centers of ω_1 and ω_2.



Proof. For the "if" direction, observe that AB and CD are the radical axes of $(ABCD), \omega_1$ and $(ABCD), \omega_2$, respectively. Thus,

$$\text{Pow}(P, \omega_1) = \text{Pow}(P, (ABCD)) = \text{Pow}(P, \omega_2),$$

which means that P must lie on the radical axis of ω_1 and ω_2 as desired.

For the "only if" direction, we know that

$$\pm PA \cdot PB = \text{Pow}(P, \omega_1) = \text{Pow}(P, \omega_2) = \pm PC \cdot PD.$$

Since $PA \cdot PB$ is negative if and only if P is between A and B and positive if and only if P is outside of segment AB, the position of P allows the converse of PoP to be satisfied. $\square$

Remark 2.7. For the case where $ABCD$'s circumcenter does lie on the line connecting the centers of ω_1 and ω_2, the radical axes are either all the same line or pairwise parallel (as they are all perpendicular to the line connecting all three centers), so they intersect at infinity. This is an important potential configuration issue that should always be remembered when using the radical center!

Now let's do some examples.

Example 2.8

Let two circles ω_1 and ω_2 meet at A, B, and let C, D be points on ω_1, ω_2, respectively, such that CD is a common tangent. Then if M is the midpoint of CD, we have

$$\mathrm{Pow}(M, \omega_1) = CM^2 = DM^2 = \mathrm{Pow}(M, \omega_2) = CM \cdot DM = \mathrm{Pow}(M, (ACD)),$$

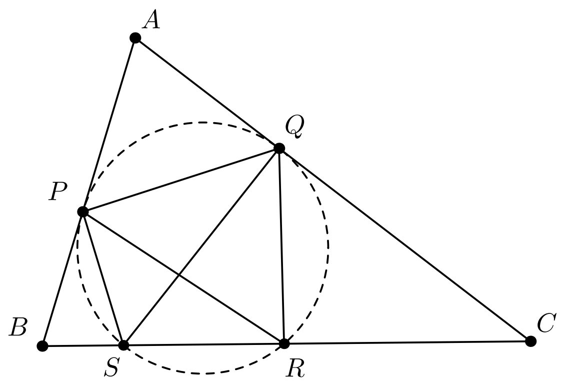
so why isn't M the radical center of ω_1, ω_2, and (ACD)?

Proof. The problem is that there are sign issues. The powers of M with respect to ω_1 and ω_2 are positive (note that M still lies on their radical axis), but the power of M with respect to (ACD) is negative. $\square$

Remark 2.9. If two circles are non-intersecting, the midpoints of the internal tangents also lie on the radical axis. This is an important property of the radical axis, so don't forget it!

Example 2.10 (USAJMO 2012/1)

Given a triangle ABC, let P and Q be points on segments AB and AC, respectively, such that $AP = AQ$. Let S and R be distinct points on segment BC such that S lies between B and R, $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R and S are concyclic



Proof. First, we observe that AB and AC are tangent to (PSR) and (QSR), respectively, and we want to show that these two circles must actually be the same. Well, if $(PSR) \neq (QSR)$, then they must have radical axis SR. Thus, if we were to show that a point not lying on SR must be on the radical axis of the two circles, we would be done by contradiction, and in fact A does the trick:

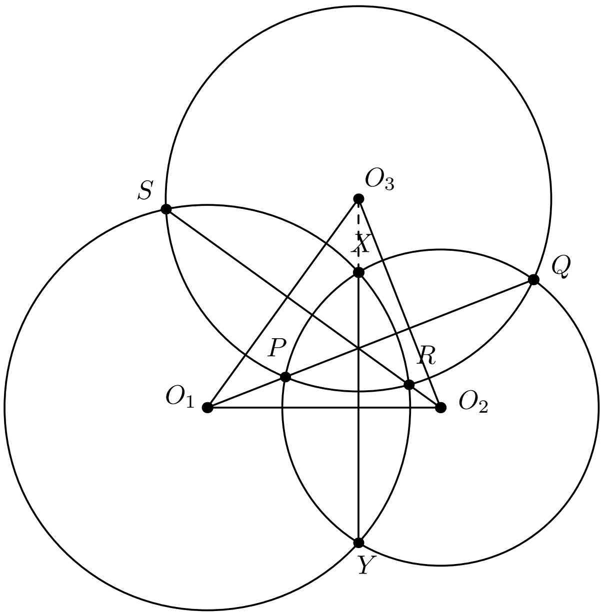
$$\mathrm{Pow}(A, (PSR)) = AP^2 = AQ^2 = \mathrm{Pow}(A, (QSR)),$$

so P, Q, S, R are concyclic as desired. $\square$

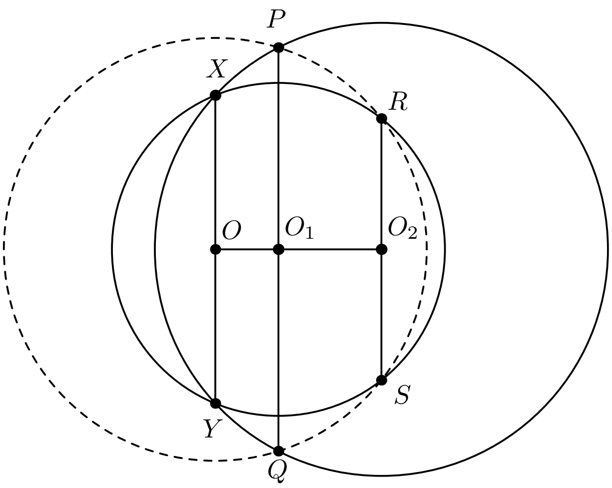
Exercise 2.11. Try this problem with phantom point instead of radical axis.

Example 2.12 (USAMO 2009/1)

Given circles ω_1 and ω_2 intersecting at points X and Y, let ℓ_1 be a line through the center of ω_1 intersecting ω_2 at points P and Q and let ℓ_2 be a line through the center of ω_2 intersecting ω_1 at points R and S. Prove that if P, Q, R and S lie on a circle then the center of this circle lies on line XY.



Proof. Let O_1, O_2, O_3 be the centers of $\omega_1, \omega_2, (PRQS)$, respectively. Drawing the diagram, we can see that by radical center, PQ and RS meet on XY, so we would like to somehow tie this information to the other condition that the radical axes pass through the centers of ω_1 and ω_2. Well, we know that $O_1Q \perp O_2O_3$ and $O_2S \perp O_1O_3$, so in fact this implies that the radical center of the three circles is actually the orthocenter of $\triangle O_1O_2O_3$. We also know that $O_2O_3 \perp XY$, so O_3, X, Y must be collinear as the three points form the altitude from O_3 to O_1O_2. However, we must remember to check whether or not O_1, O_2, O_3 can be collinear! In fact, as seen in the diagram below, this is actually a possible configuration:



Thankfully, this special case can just be addressed with a fairly simple phantom point and Pythagorean Theorem argument: if we let O be the midpoint of XY, then

$$\begin{aligned} OP^2 &= OO_1^2 + O_1P^2 \\ &= OO_1^2 + (O_2P^2 - O_1O_2^2) \\ &= OO_1^2 + O_2X^2 - O_1O_2^2 \\ &= OO_1^2 + (OO_2^2 + OX^2) - O_1O_2^2 \\ &= OO_1^2 + OO_2^2 + \left(\frac{1}{2}XY\right)^2 - O_1O_2^2, \end{aligned}$$

which is symmetrically equal for OQ^2, OR^2, OS^2 as well, so $O = O_3$. $\square$

Exercise 2.13. Prove that O_3 lies on the radical axis of ω_1, ω_2 with length chasing rather than radical center. Note that this solution does not require the all parallel case to be addressed separately!

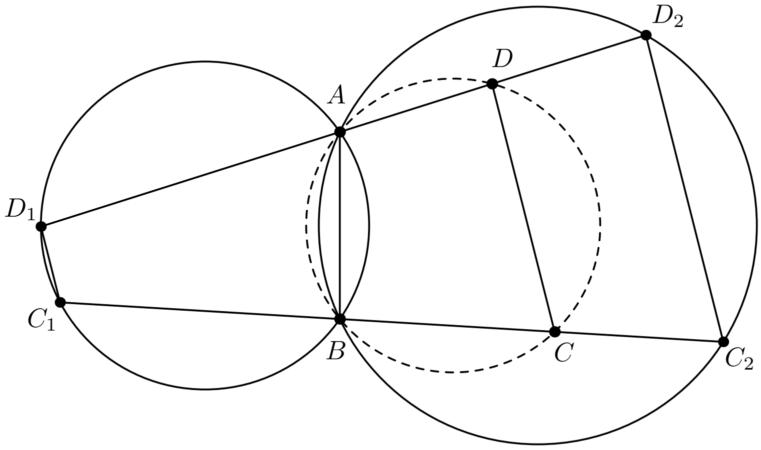
§3 Coaxial Circles

Definition 3.1. If 3 or more circles share pairwise radical axes, then we say that they are a **coaxial pencil** or just **coaxial**.

Note that three circles that share a point are coaxial if and only if their centers are collinear; thus, coaxiality can come in handy if we wish to prove that three circles' centers are collinear. However, the main result of this section is the following relatively obscure lemma (though it has recently become more popular), which is a very powerful tool for proving concyclicity as it allows us to essentially ignore two of the points on the circle.

Lemma 3.2 (Forgotten Coaxiality Lemma)

Let two circles ω_1 and ω_2 intersect at points A and B. Then the locus of points P such that $\frac{\text{Pow}(P,\omega_1)}{\text{Pow}(P,\omega_2)} = k$ for some constant k is a circle that forms a coaxial pencil with ω_1 and ω_2.



Proof. WLOG let k be negative (i.e. the circle is contained within the union of the areas of ω_1 and ω_2). Let C and D be points such that $k = \frac{\text{Pow}(C,\omega_1)}{\text{Pow}(C,\omega_2)} = \frac{\text{Pow}(D,\omega_1)}{\text{Pow}(D,\omega_2)}$. If AD meets ω_1, ω_2 at D_1, D_2, respectively, and BC meets ω_1, ω_2 at C_1, C_2, respectively, then it follows that

$$k = -\frac{CC_1 \cdot CB}{CC_2 \cdot CB} = -\frac{DD_1 \cdot DA}{DD_2 \cdot DA} \implies \frac{CC_1}{CC_2} = \frac{DD_1}{DD_2};$$

along with $C_1D_1 \parallel C_2D_2$ from Reim's Theorem on ABC_1D_1, this gives us $CD \parallel C_1D_1 \parallel C_2D_2$, which means that $ABCD$ is cyclic by the converse of Reim. These steps can be reversed to prove the converse; assuming $ABCD$ is cyclic, Reim's implies that $CD \parallel C_1D_1 \parallel C_2D_2$, which means that

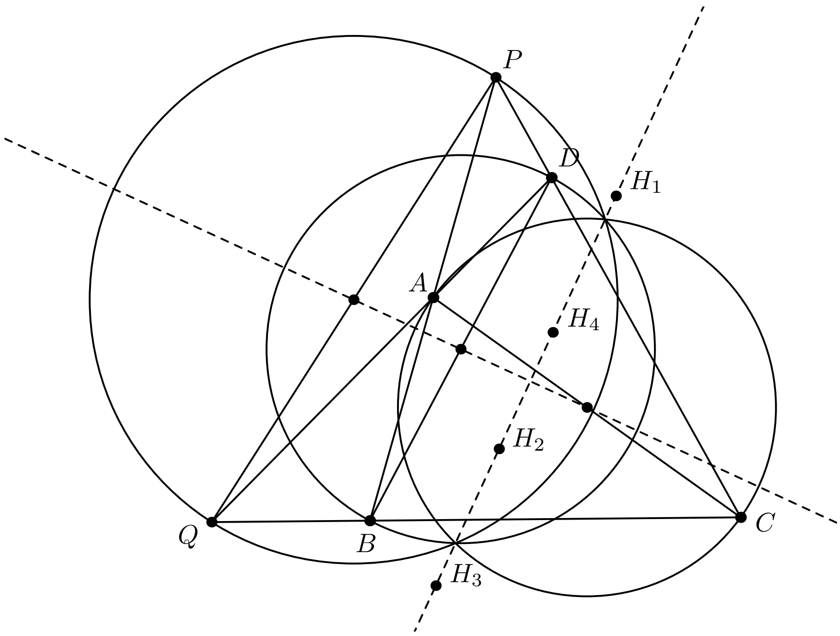
$$\frac{CC_1}{CC_2} = \frac{DD_1}{DD_2} \implies \frac{CC_1 \cdot CB}{CC_2 \cdot CB} = \frac{DD_1 \cdot DA}{DD_2 \cdot DA}$$

as desired. □

First, we present two classical results.

Example 3.3 (Newton-Gauss Line)

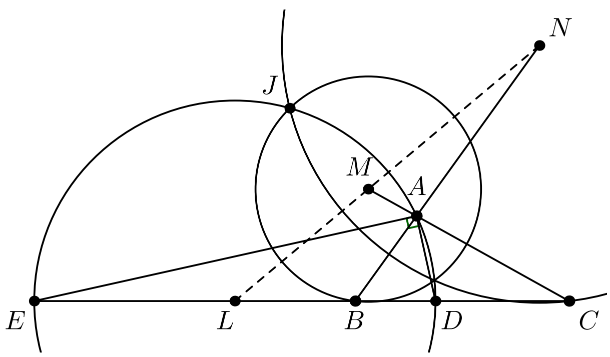
In a convex quadrilateral $ABCD$, let $P = AB \cap CD$ and $Q = AD \cap BC$. Then the midpoints of PQ, AC, and BD are collinear.



Proof. We will show that the circles $(PQ), (AC)$, and (BD) are coaxial, which suffices. Let H_1, H_2, H_3, and H_4 be the orthocenters of $\triangle PAD, \triangle PBC, \triangle QAB$, and $\triangle QCD$, respectively. Then since H_1 is the radical center of the triples $\{(PQ), (BD), (PD)\}$, $\{(PQ), (AC), (AP)\}$, and $\{(BD), (AC), (AD)\}$, it lies on the pairwise radical axes of $(PQ), (AC)$, and (BD). Similarly, H_2, H_3, and H_4 also lie on the pairwise radical axes, which means that the three circles must actually be coaxial as they would otherwise have multiple radical centers, an impossibility. $\square$

Example 3.4 (Existence of Isodynamic Points)

In a triangle $\triangle ABC$, the locus of points P such that $\frac{PB}{PC} = \frac{AB}{AC}$ is known as the A-**Apollonian Circle**. Prove that the three Apollonian Circles of a non-equilateral triangle are coaxial.



Proof. First we show that the locus is actually a circle. To do so, let D and E be the intersections of the A-internal and A-external bisectors of $\triangle ABC$ with BC, respectively; observe that D and E must lie on the A-Apollonian Circle because of the Angle Bisector Theorem on $\triangle ABC$, and we also must have $\angle DAE = 90°$. This suggests that the locus is the circle with diameter DE; indeed, for any other point P satisfying $\frac{PB}{PC} = \frac{AB}{AC}$, D and E are also the intersections of the P-internal and P-external bisectors of $\triangle PBC$ with BC, so again we must have $\angle DPE = 90°$ as desired.

Next we show that the three Apollonian Circles are coaxial. Let J be an intersection of the A- and B-Apollonian Circles of $\triangle ABC$. Then we must have

$$\frac{JB}{JC} = \frac{AB}{AC}, \frac{JA}{JC} = \frac{BA}{BC} \implies \frac{JB}{JC} = \frac{AC}{BC},$$

so J lies on the C-Apollonian Circle as well. Let L, M, N be the centers of the A-, B-, C-Apollonian Circles, respectively; it now suffices to show that L, M, N are collinear. Since L is the midpoint of DE, we have

$$\angle LAB + \frac{1}{2}\angle A = \angle ADL = \frac{1}{2}\angle A + \angle C \implies \angle LAB = \angle C,$$

so LA is tangent to (ABC), and by symmetry MB and NC are as well, which means that

$$\triangle LAB \sim \triangle LCA \implies \frac{BC^2}{CA^2} = \frac{LA^2}{LC^2} = \frac{LB \cdot LC}{LC^2} = \frac{LB}{LC}.$$

Thus,

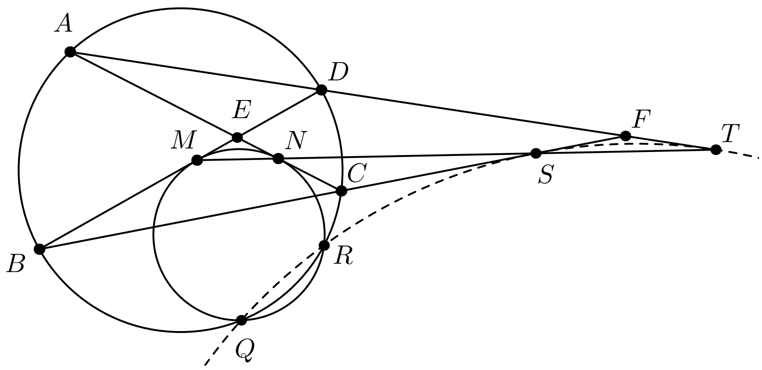
$$\frac{LB}{LC} \cdot \frac{MC}{MA} \cdot \frac{NA}{NB} = \frac{BC^2}{CA^2} \cdot \frac{CA^2}{AB^2} \cdot \frac{AB^2}{BC^2} = 1,$$

so we're done by the converse of Menelaus's Theorem on $\triangle ABC$ with traversal LMN. □

Now we obliterate some problems with the Forgotten Coaxiality Lemma.

Example 3.5 (CGMO 2017/7)

Let the $ABCD$ be a cyclic quadrilateral with circumcircle ω_1. Lines AC and BD intersect at point E, and lines AD, BC intersect at point F. Circle ω_2 is tangent to segments EB, EC at points M, N respectively, and intersects with circle ω_1 at points Q, R. Lines BC, AD intersect line MN at S, T respectively. Show that Q, R, S, T are concyclic.



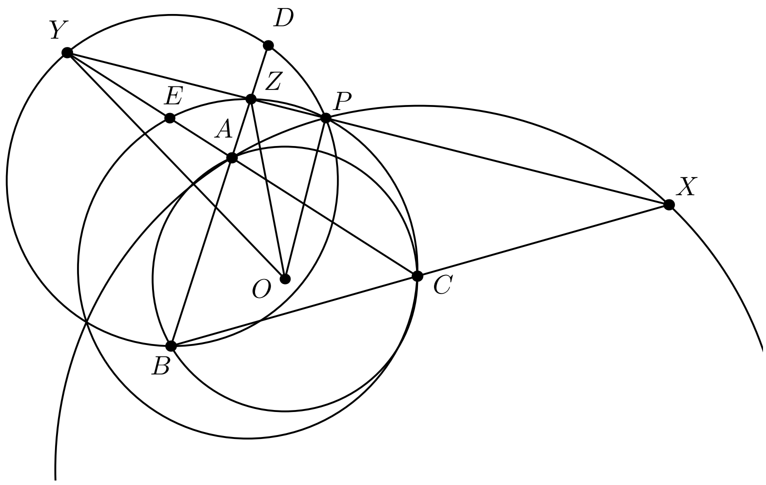
Proof. We're motivated to use the Forgotten Coaxiality Lemma because there are lots of similar triangles, so we know a lot of lengths, and there's not really any obvious way to characterize Q and R. Through some easy angle chasing, we can observe that $\triangle SCN \sim \triangle TDM \implies \frac{SN}{TM} = \frac{SC}{TD}$ and $\triangle SBM \sim \triangle TAN \implies \frac{SM}{TN} = \frac{SB}{TA}$, so it follows that

$$\frac{SC \cdot SB}{SN \cdot SM} = \frac{TD \cdot TA}{TN \cdot TM} \iff \frac{\text{Pow}(S, \omega_1)}{\text{Pow}(S, \omega_2)} = \frac{\text{Pow}(T, \omega_1)}{\text{Pow}(T, \omega_2)},$$

and we're done. □

Example 3.6 (IMO Shortlist 2012/G8)

Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω. Denote by P the foot of the perpendicular from the center of ω to ℓ. The side-lines BC, CA, AB intersect ℓ at the points X, Y, Z different from P. Prove that the circumcircles of the triangles AXP, BYP and CZP have a common point different from P or are mutually tangent at P.



Proof. WLOG suppose the diagram is as shown, and let $D = AB \cap (BYP) \neq B$ and $E = AC \cap (CZP) \neq C$. By the Forgotten Coaxiality Lemma, we just need to show that

$$\frac{\mathrm{Pow}(A,(BYP))}{\mathrm{Pow}(A,(CZP))} = \frac{\mathrm{Pow}(X,(BYP))}{\mathrm{Pow}(X,(CZP))} \iff \frac{AD \cdot AB}{AC \cdot AE} = \frac{XP \cdot XY}{XP \cdot XZ}.$$

By Menelaus's Theorem on $\triangle AYZ$ and traversal BCX, we find that the RHS is equal to $\frac{AB \cdot CY}{BZ \cdot AC}$, so it suffices to prove that $\frac{AD}{AE} = \frac{CY}{BZ}$. Now this is just length chasing: we have

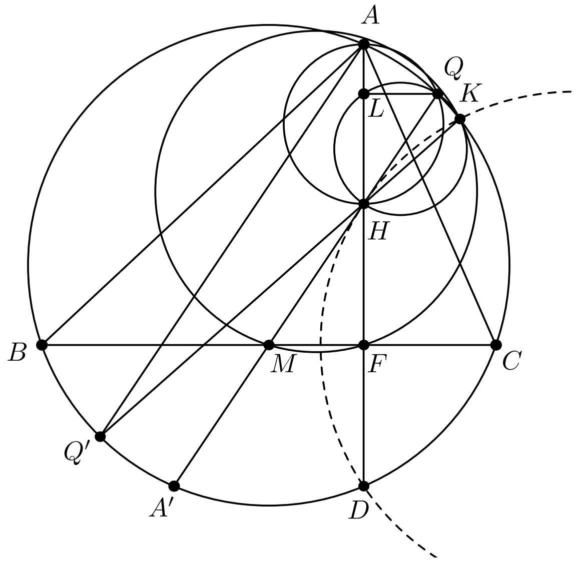
$$\begin{aligned}
&AD \cdot BZ - AE \cdot CY + ZP \cdot ZY - YP \cdot ZY \\
&= AD \cdot BZ - AE \cdot CY + DZ \cdot BZ - EY \cdot CY \\
&= AZ \cdot BZ - AY \cdot CY \\
&= \mathrm{Pow}(Z,(ABC)) - \mathrm{Pow}(Y,(ABC)) \\
&= OZ^2 - OY^2 \\
&= (OP^2 + ZP^2) - (OP^2 + YP^2) \\
&= ZY(ZP - YP),
\end{aligned}$$

so we're done upon rearranging. □

For our final example, we will use the Forgotten Coaxiality Lemma on a circle with radius 0(!) to prove that two circles are tangent.

Example 3.7 (IMO 2015/3)

Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocenter, and F the foot of the altitude from A. Let M be the midpoint of BC. Let Q be the point on Γ such that $\angle HQA = 90^\circ$ and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K and Q are all different and lie on Γ in this order. Prove that the circumcircles of triangles KQH and FKM are tangent to each other.



Proof. Let $L = Q\infty_{BC} \cap AH$. The problem is equivalent to showing that $(KQH), (FKM)$, and the circle centered at K with radius 0 are coaxial, so by the Forgotten Coaxiality Lemma we need to prove that

$$\frac{\text{Pow}(M,K)}{\text{Pow}(M,(KQH))} = \frac{\text{Pow}(F,K)}{\text{Pow}(F,(KQH))} \iff \frac{MK^2}{FK^2} = \frac{MH \cdot MQ}{FH \cdot FL}.$$

However, $\triangle HMF \overset{+}{\sim} \triangle HQL$ implies that $\frac{MH}{FH} = \frac{QH}{LH}$, so $\frac{MQ}{FL} = \frac{MH+QH}{FH+LH} = \frac{MH}{FH}$, which means that we just want to show that $\frac{MK}{FK} = \frac{MH}{FH}$; this is equivalent to showing that K lies on ω, the H-Apollonian Circle of $\triangle HMF$. If P is the center of ω and D is the reflection of H over F, then it's clear that D must lie on the ω as $DF = HF$ and $DM = HM$, and we also know that PH is tangent to (HMF), so HM is tangent to ω. Thus, it suffices to show that HM is tangent to (DHK).

Let A' be the reflection of H over M, ℓ' be the tangent to (DKH) at H, and $KH \cap \Gamma \neq K = Q'$; clearly Q, H, M, A' are collinear because $\angle AQH = 90^\circ = \angle AQA'$, and it's well-known that A' is the A-antipode in Γ. Additionally, Q' must be the antipode of Q in (ABC) because $\angle QKQ' = \angle QKH = 90^\circ$, implying that $AQA'Q'$ is a rectangle, which combined with $\ell' \parallel AQ'$ from Reim's Theorem on degenerate quadrilateral $HHKD$ gives $HM = \ell'$ as desired. $\square$

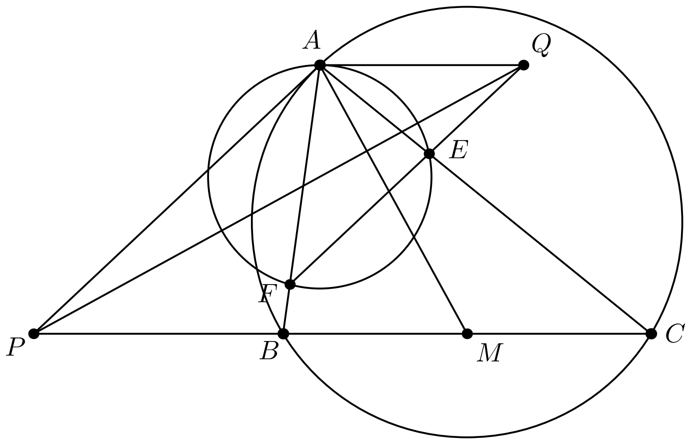
§4 Point Circles

We now take a brief moment to examine the uses of the radical axis when we consider points to be circles with radius 0. Although this technique is fairly niche and extremely difficult to spot in the wild, it often leads to very elegant solutions when used effectively.

Example 4.1 (Circumcenter)
In $\triangle ABC$, prove that the perpendicular bisectors of its sides meet at a single point.

Proof. If we consider A and B to be circles of radius 0, then it follows that their radical axis is the perpendicular bisector of AB. Symmetrically, it follows that the radical center of A, B, C is the concurrent point of the three perpendicular bisectors. □

Example 4.2 (Polish MO 2018/5)
An acute triangle ABC in which $AB < AC$ is given. Points E and F are feet of its heights from B and C, respectively. The line tangent in point A to the circle escribed on ABC crosses BC at P. The line parallel to BC that goes through point A crosses EF at Q. Prove PQ is perpendicular to the median from A of triangle ABC.

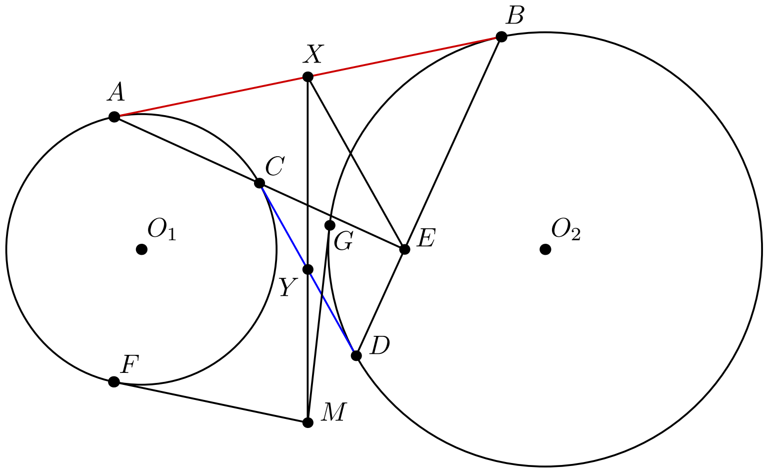


Proof. We observe that if we were to show that PQ was the radical axis of (BC) and some circle centered at A, we would be done, which motivates our next step: since EF is the radical axis of (BC) and (AEF), it follows that Q is the radical center of the point circle at A, so Q lies on the radical axis of A and (BC), and obviously we have

$$\operatorname{Pow}(P, A) = PA^2 = \operatorname{Pow}(P, (ABC)) = PB \cdot PC = \operatorname{Pow}(P, (BC))$$

so P lies on the radical axis of A and (BC) as well. □

Example 4.3 (CGMO 2015/6)
Let Γ_1 and Γ_2 be two non-overlapping circles. A, C are on Γ_1 and B, D are on Γ_2 such that AB is an external common tangent to the two circles, and CD is an internal common tangent to the two circles. AC and BD meet at E. F is a point on Γ_1, the tangent line to Γ_1 at F meets the perpendicular bisector of EF at M. MG is a line tangent to Γ_2 at G. Prove that $MF = MG$.



Proof. We observe that there are a lot of equal lengths in this problem, so this motivates us to try to use radical axes somehow. Since we know that the midpoints of AB and CD form the radical axis of Γ_1 and Γ_2 (see Example 2.8 if you're not sure why), we let these points be X and Y, respectively. We also know that $AC \perp BD$ through some simple angle chasing, so we now introduce the key idea of using the point circle E :

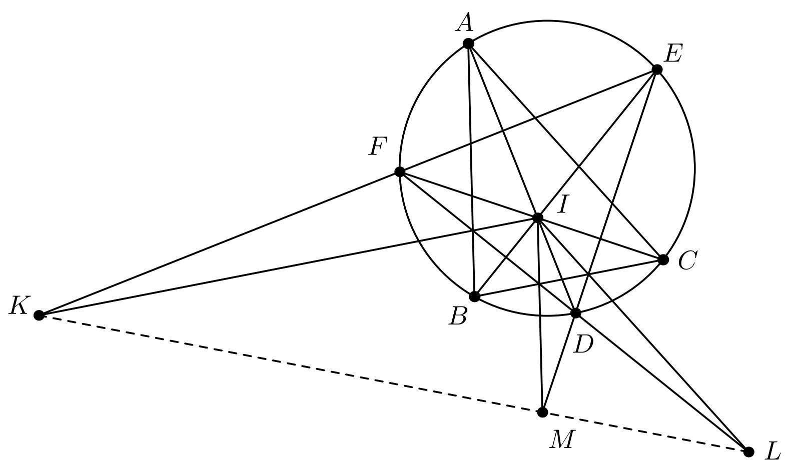
$$\text{Pow}(X, E) = XE^2 = XA^2 = \text{Pow}(X, A) = XB^2 = \text{Pow}(X, B)$$
$$\text{Pow}(Y, E) = YE^2 = YA^2 = \text{Pow}(Y, A) = YB^2 = \text{Pow}(Y, B),$$

implying that Γ_1, Γ_2, and E are coaxial. Thus, $\text{Pow}(M, \Gamma_1) = MF^2 = ME^2 = \text{Pow}(M, E)$ implies that $M \in XY$ as well, which means that $MF^2 = MG^2 \implies MF = MG$. □

This next solution is in my opinion less intuitive than the previous ones, but it's too good not to share.

Example 4.4 (Balkan MO 2015/2)

Let $\triangle ABC$ be a scalene triangle with incentre I and circumcircle ω. Lines AI, BI, CI intersect ω for the second time at points D, E, F, respectively. The parallel lines from I to the sides BC, AC, AB intersect EF, DF, DE at points K, L, M, respectively. Prove that the points K, L, M are collinear.



Proof. Since $\angle FIK = \angle FCB = \angle KEI$, we have $\triangle KFI \stackrel{-}{\sim} \triangle KIE$. This means that $KF \cdot KE = KI^2$, so K lies on the radical axis of I and ω, and by symmetry M and L do too, which means that K, M, L are collinear as desired. □

§5 Black Magic

The main idea of this section is to combine Power of a Point with more analytical methods, which tends to be more versatile than just using PoP by itself albeit at the cost of generally being more computationally intensive.

§5.1 Barycentric Coordinates

The use of barycentric coordinates concurrently with Power of a Point is actually fairly natural due to the following fact:

Theorem 5.1 (Barycentric Power of a Point)

Let some circle ω have equation

$$-a^2yz - b^2zx - c^2xy + (x+y+z)(ux+vy+wz) = 0.$$

Then if $P = (x, y, z)$ is any point in the plane,

$$\operatorname{Pow}(P, \omega) = -a^2yz - b^2zx - c^2xy + (x+y+z)(ux+vy+wz).$$

Proof. Since the formula of a circle is derived from

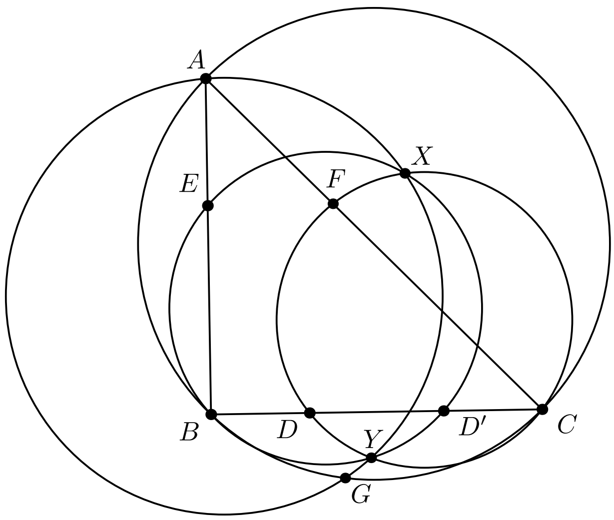
$$(\text{distance from } (x, y, z) \text{ to the center})^2 - \text{radius}^2 = 0,$$

plugging in an arbitrary point yields the power of that point. Note that (x, y, z) cannot be homogenized, because otherwise the distance formula breaks! □

As a demonstration of how useful this formula can be even by itself, we present an extremely clean solution to a problem from the AoPS Quarantine Geometry Olympiad.

Example 5.2 (AQGO 2020/2)

Let $\triangle ABC$ be a triangle with D as a random point on $\overline{BC}$ such that $\overline{BD} < \overline{BM}$ where M is the midpoint of $\overline{BC}$ and let E, F be on sides $\overline{AB}, \overline{AC}$ such that $\overline{EF} \parallel \overline{BC}$. Let D' be the reflection of D across the midpoint of $\overline{BC}$ and define X, Y as the intersections of $\odot(BD'E)$ and $\odot(CDF)$. Then prove that $\odot(AXY)$ passes through a fixed point as D varies other than A.



Proof. Upon inspection, the fixed point appears to be the intersection of the A-symmedian with (ABC); call this point G. Since D is a variable point, it's hard to get any information about it, so we are motivated to use the Forgotten Coaxiality Lemma in order to avoid having to deal with it. We use barycentric coordinates with reference triangle $\triangle ABC$. Let $E = (e, 1-e, 0)$ and $F = (e, 0, 1-e)$ (this comes from $\frac{[EBC]}{[ABC]} = \frac{[FBC]}{[ABC]}$) be fixed points, and let $D = (0, d, 1-d)$ and $D' = (0, 1-d, d)$ be variable points. Plugging in B, E, D' and C, F, D implies that (BED') and (CFD) are parametrized by

$$a^2yz + b^2zx + c^2xy = (x+y+z)(c^2(1-e)x + a^2(1-d)z)$$

and

$$a^2yz + b^2zx + c^2xy = (x+y+z)(b^2(1-e)x + a^2(1-d)y),$$

respectively. Since $G = (-a^2 : 2b^2 : 2c^2)$, it follows that

$$\begin{aligned}
&\frac{\text{Pow}_{(BED')}(G)}{\text{Pow}_{(CFD)}(G)} \\
&= \frac{-4a^2b^2c^2 + 2a^2b^2c^2 + 2a^2b^2c^2 + (2b^2+2c^2-a^2)(-a^2c^2(1-e) + 2a^2c^2(1-d))}{-4a^2b^2c^2 + 2a^2b^2c^2 + 2a^2b^2c^2 + (2b^2+2c^2-a^2)(-a^2b^2(1-e) + 2a^2b^2(1-d))} \\
&= \frac{c^2}{b^2} \\
&= \frac{AE \cdot AB}{AF \cdot AC} \\
&= \frac{\text{Pow}_{(BED')}(A)}{\text{Pow}_{(CFD)}(A)},
\end{aligned}$$

so A, G, X, Y are concyclic, which is what we wanted to show. Note that we were able to use homogenized coordinates here because the common denominator of unhomogenized coordinates would cancel out in the fraction. This will not always be the case! □

Our next example is an occasionally useful result that allows us to compute arbitrary powers of points with respect to a circle without having to involve its center.

Example 5.3

Let ABC be a triangle and P be an arbitary point. Let the parallel line to BC through P meet AB, AC at A_1, A_2. Similarly define $B_1 \in BC, B_2 \in BA$ and $C_1 \in CA, C_2 \in CB$. Then using directed lengths, i.e. $PA_1 \cdot PA_2$ is given a positive sign if $\vec{PA_1}$ is in the same direction of $\vec{PA_2}$ and negative otherwise, we have

$$\text{Pow}(P, (ABC)) = PA_1 \cdot PA_2 + PB_1 \cdot PB_2 + PC_1 \cdot PC_2.$$

Proof. We use barycentric coordinates with reference triangle $\triangle ABC$. Suppose that $P = (x, y, z)$. Observing that $[PBC] = [A_1BC] = [A_2BC]$ since the three triangles share the same base and height, we must have $A_1 = (x, 1-x, 0), A_2 = (x, 0, 1-x)$, so the barycentric distance formula implies that

$$PA_1 = \sqrt{-a^2(x+y-1)z} = az, PA_2 = ay,$$

and similar computations yield

$$PB_1 = bx, PB_2 = bz, PC_1 = cy, PC_2 = cx.$$

Note that if the directed length product $PA_1 \cdot PA_2$ is positive, then P is outside of $\triangle ABC$ and one of y, z must be negative, and if $PA_1 \cdot PA_2$ is negative, then P is inside $\triangle ABC$ and both y and z must be positive. Thus, we must have $PA_1 \cdot PA_2 = -a^2yz$ in order for both sides to have the same sign, so we get

$$\begin{aligned}&PA_1 \cdot PA_2 + PB_1 \cdot PB_2 + PC_1 \cdot PC_2\\&= -a^2yz + -b^2zx + -c^2xy\\&= \operatorname{Pow}(P, (ABC))\end{aligned}$$

as desired. □

§5.2 Linearity

We begin with Remark 2.5 from the second section, but formalized:

Theorem 5.4 (Linearity of PoP)

Let ω_1 and ω_2 be two circles and $\mathcal{P}$ be the set of points in the plane. Then the function $f : \mathcal{P} \to \mathbb{R}$ defined by $f(\bullet) = \operatorname{Pow}(\bullet, \omega_1) - \operatorname{Pow}(\bullet, \omega_2)$ is linear; in particular, if $\vec{P} = t\vec{A} + (1-t)\vec{B}$, then $f(P) = tf(A) + (1-t)f(B)$.

Proof. We already proved this earlier, but we now present an alternate, more synthetic proof. Let A, B, C be points such that $\vec{C} = k\vec{A} + (1-k)\vec{B}$. If O_1 and O_2 are the centers of ω_1 and ω_2, respectively, then Stewart's Theorem on $\triangle O_1AB, \triangle O_2AB$ respectively imply that

$$k(1-k) + CO_1^2 = kAO_1^2 + (1-k)BO_1^2,$$
$$k(1-k) + CO_2^2 = kAO_2^2 + (1-k)BO_2^2.$$

Subtracting the second equality from the first, we have

$$\begin{aligned}CO_1^2 - CO_2^2 &= k(AO_1^2 - AO_2^2) + (1-k)(BO_1^2 - BO_2^2)\\CO_1^2 - r_1^2 - CO_2^2 + r_2^2 &= k(AO_1^2 - r_1^2 - AO_2^2 + r_2^2) + (1-k)(BO_1^2 - r_1^2 - BO_2^2 + r_2^2)\\f(C) &= kf(A) + (1-k)f(B)\end{aligned}$$

as desired. □

The way that this theorem is mostly used in practice is the following corollary:

Corollary 5.5

Let ω_1, ω_2 be two circles and let P be a point on line BC. Then if $f(\bullet) = f(\bullet, \omega_1) - f(\bullet, \omega_2)$, we have

$$f(P) = \frac{PC}{BC}f(B) + \frac{PB}{BC}f(C).$$

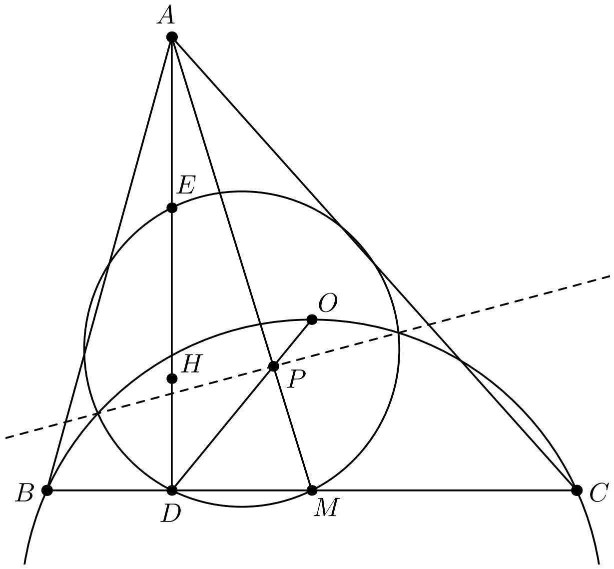
More generally, if $P = (x, y, z)$ in barycentric coordinates with reference triangle $\triangle ABC$, we have

$$f(P) = xf(A) + yf(B) + zf(C).$$

Now let's do some examples, first with the line form of Corollary 5.5, and then with the general form.

Example 5.6

In $\triangle ABC$, let M be the midpoint of BC, D be the projection of A onto BC, and O be the circumcenter. OD meets AM at P. Prove that P lies on the radical axis of (BOC) and the nine-point circle of triangle ABC.



Proof. WLOG let $\triangle ABC$ be acute. Let ω be the nine-point circle, and define $f(\bullet) = \operatorname{Pow}(\bullet, (BOC)) - \operatorname{Pow}(\bullet, \omega)$. Here, we actually have two choices for $f(P)$, since

$$\vec{P} = \frac{OP}{OD}\vec{D} + \frac{DP}{OD}\vec{O} = \frac{PM}{AM}\vec{A} + \frac{AP}{AM}\vec{M}.$$

We will present the former solution path and leave the latter as an exercise. Let H be the orthocenter of $\triangle ABC$, and let AD meet ω again at E (well-known to be the midpoint of AH). Observe that it suffices to show that

$$0 = f(P) = \frac{OP}{OD}f(D) + \frac{DP}{OD}f(O) \iff \frac{OP}{DP} = -\frac{f(O)}{f(D)}.$$

It's well-known that O and H are reflections over the center of ω, so we actually have

$$\begin{aligned} &f(O) = -\operatorname{Pow}(O,\omega) = -\operatorname{Pow}(H,\omega) = EH \cdot HD = (R\cos\angle A)(2R\cos\angle B\cos\angle C), \\ &f(D) = \operatorname{Pow}(D,(BOC)) = -BD \cdot DC = -(AB\cos\angle B)(AC\cos\angle C), \\ &\implies -\frac{f(O)}{f(D)} = \frac{2R^2\cos\angle A}{AB \cdot AC}. \end{aligned}$$

We also know that $OM \parallel AD$, so

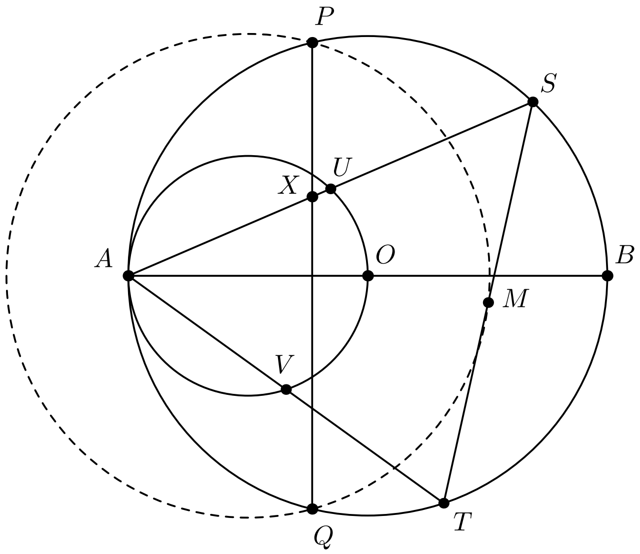
$$\triangle OPM \sim \triangle DPA \implies \frac{OP}{DP} = \frac{OM}{DA} = \frac{R\cos\angle A}{AB\sin\angle B} = \frac{2R^2\cos\angle A}{AB \cdot AC},$$

where the last equality comes from the Law of Sines on $\triangle ABC$, as desired. □

Exercise 5.7. Prove that $\frac{AP}{MP} = \frac{f(A)}{f(M)}$.

Example 5.8 (USAMO 2015/2)

Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.



Proof. Let O be the center of ω. In locus problems, it's always a good idea to check the extreme cases, so setting $X = P, X = PQ \cap AB$, and $X = Q$, we can see that M lies on the circle passing through the midpoint of PB, the midpoint of QB, and Q. Thus, we predict that the center of the locus circle of M is the midpoint of AO. This motivates us to construct the circle with diameter AO as well as the midpoints U and V of AS and AT, respectively, which clearly lie on (AO). Now let $f(\bullet) = \text{Pow}(\bullet, (AO)) - \text{Pow}(\bullet, \omega)$; then we have

$$\text{Pow}(M, (AO)) + MS \cdot MT = f(M) = \frac{1}{2}(f(S) + f(T)) = \frac{1}{2}(SU \cdot SA + VT \cdot AT),$$

which implies that $\text{Pow}(M, (AO)) = \frac{1}{4}(SA^2 + AT^2 - TS^2)$. By Law of Cosines on $\triangle AST$, this quantity equals

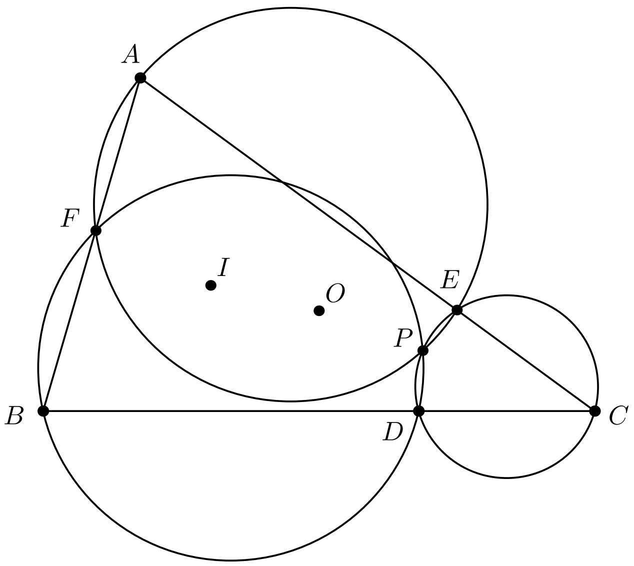
$$\frac{1}{2}(SA \cdot AT \cos \angle SAT) = \frac{1}{2}(SA \cdot AX),$$

but we know that $\triangle APX \sim \triangle ASP \implies SA \cdot AX = AP^2$, so $\text{Pow}(M, (AO)) = \frac{1}{2}AP^2$. In particular, this quantity does not depend on X, so if O_1 is the midpoint of AO, it follows that MO_1^2 is fixed from the definition of power of a point, meaning that M lies on a fixed circle with center O_1 as desired. $\square$

Like the Balkan problem in the previous section, this next solution is definitely far less natural than the previous ones, but it's extremely slick.

Example 5.9 (IMO Shortlist 2012/G6)

Let ABC be a triangle with circumcenter O and incenter I. The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.



Proof. First, it's clear that $AE + AF = BC$, and we also observe that P lies on (AEF) by Miquel's Pivot Theorem. It suffices to show that $\text{Pow}(P, (ABC)) = \text{Pow}(I, (ABC))$, which happens if and only if $\text{Pow}(P, (ABC)) = -2Rr$ by Euler's Theorem. Let P have barycentric coordinates (x, y, z) with reference triangle $\triangle ABC$. Then if we define

$$\begin{aligned} f_A(\bullet) &= \text{Pow}(\bullet, (AEF)) - \text{Pow}(\bullet, (ABC)), \\ f_B(\bullet) &= \text{Pow}(\bullet, (BFD)) - \text{Pow}(\bullet, (ABC)), \\ f_C(\bullet) &= \text{Pow}(\bullet, (CDE)) - \text{Pow}(\bullet, (ABC)), \end{aligned}$$

we have

$$\begin{aligned} -\text{Pow}(P, (ABC)) &= xf_A(A) + yf_A(B) + zf_A(C) = y(c \cdot BF) + z(b \cdot CE) \\ &= xf_B(A) + yf_B(B) + zf_B(C) = z(a \cdot CD) + x(c \cdot AF) \\ &= xf_C(A) + yf_C(B) + zf_C(C) = x(b \cdot AE) + y(a \cdot BD), \end{aligned}$$

so

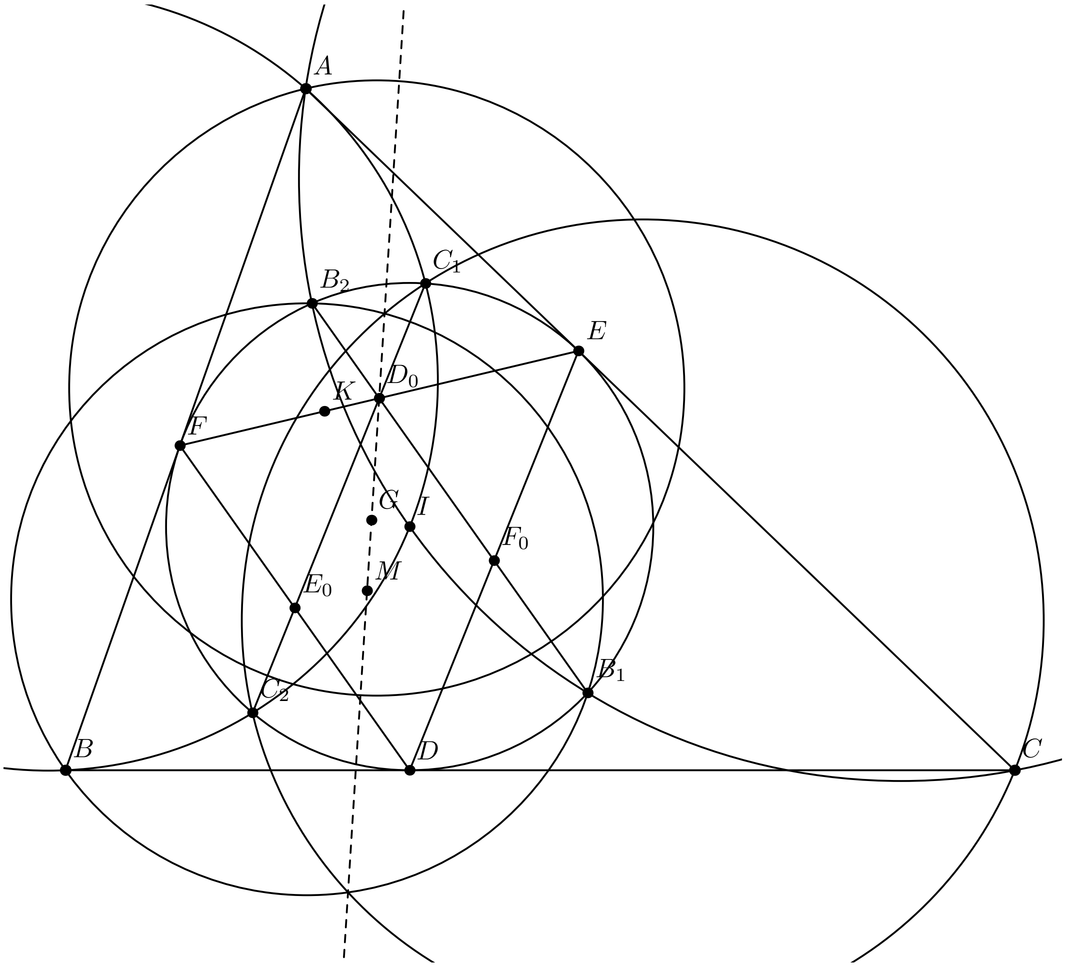
$$-\text{Pow}(P, (ABC)) \sum_{cyc} a = \sum_{cyc} yca(BF) + zba(CE) = \sum_{cyc} xbc(AE + AF) = abc \sum_{cyc} x = abc.$$

Thus, $\text{Pow}(P, (ABC)) = -\frac{abc}{a+b+c} = -\frac{4R[ABC]}{\frac{2[ABC]}{r}} = -2Rr$ as desired. □

Now to finish, we will brutally murder a classic problem to showcase how powerful linearity can be.

Example 5.10 (TSTST 2016/6)

Let ABC be a triangle with incenter I, and whose incircle is tangent to $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ at D, E, F, respectively. Let K be the foot of the altitude from D to $\overline{EF}$. Suppose that the circumcircle of $\triangle AIB$ meets the incircle at two distinct points C_1 and C_2, while the circumcircle of $\triangle AIC$ meets the incircle at two distinct points B_1 and B_2. Prove that the radical axis of the circumcircles of $\triangle BB_1B_2$ and $\triangle CC_1C_2$ passes through the midpoint M of $\overline{DK}$.



Proof. Call the incircle ω. After some messing around, we let D_0, E_0, F_0 be the midpoints of EF, FD, DE, respectively: we can observe that D_0 is the radical center of $(AIC), \omega, (AI)$ and F_0 is the radical center of $(AIC), \omega, (CI)$, so B_1B_2 is the E-midline in $\triangle DEF$ (of course, C_1C_2 is the F-midline as well by symmetry). As the Gergonne Point G of $\triangle ABC$ is the Symmedian Point of $\triangle D_0E_0F_0$, it's well-known that D_0, M, G are collinear (the so-called D-Schwatt Line of $\triangle DEF$), so now we just need to show that G lies on the radical axis of (BB_1B_C) and (CC_1C_2).

To do so, we will use linearity: define $f_B(\bullet) = \text{Pow}(\bullet, \omega) - \text{Pow}(\bullet, (BB_1B_2))$ and f_C similarly. Using $\triangle BF_0D_0$ as a reference triangle, we have

$$f_B(G) = \frac{[GF_0D_0]}{[BF_0D_0]} f_B(B) + \frac{[GBF_0]}{[BF_0D_0]} f_B(F_0) + \frac{[GD_0B]}{[BF_0D_0]} f_B(E_0) = \frac{\text{dist}(G, F_0D_0)}{\text{dist}(B, F_0D_0)} BD^2.$$

Thus, by symmetry it suffices to show that

$$\frac{f_B(G)}{f_C(G)} = 1 \iff \frac{CD^2}{BD^2} \cdot \frac{\text{dist}(G, D_0E_0)}{\text{dist}(G, F_0D_0)} \cdot \frac{\text{dist}(B, F_0D_0)}{\text{dist}(C, D_0E_0)} = 1.$$

Noting that there are many right and equal angles from the tangents in the configuration, we are motivated to use trigonometry; we complete the rest of this proof in Appendix 9.3, as it is not very enlightening and fairly long. $\square$

Exercise 5.11. For an alternate approach, let the line through G parallel to FD meet AB, BC at Q', S'. The crux of the solution is showing that $(BQ'S'), (DEF), (AIC)$ are coaxial, which can be done in many ways (see here, here, or here).

§6 Final Remarks

§6.1 Heuristics

Of course, knowing a theorem's statement doesn't mean anything unless you also know when and how to apply it. Therefore, this section is dedicated to listing some situations where Power of a Point is likely going to be useful.

- Power of a Point is first and foremost a length chasing tool. If a problem involves a circle and asks for some condition that can be encoded into length ratios, start computing powers of points!
- The orthocenter plays very nicely with Power of a Point. Thus, even when there might not be any obvious circles in a problem, if there is a prominent orthocenter, don't count Power of a Point out of the equation just yet.
- It's unlikely that the correct points to use PoP on will be immediately obvious, so don't be afraid to introduce lots of extra points until you find the right ones.
- If a problem involves proving that four points are concyclic but there's very little information about one of them, don't start blindly flailing about with the basic PoP theorem; the Forgotten Coaxiality Lemma is probably going to be more useful.
- Power of a Point is extremely powerful (no pun intended) for problems involving tangencies if the examples haven't made that obvious already. The first thing you should try should just be the basic PoP theorem, but don't be afraid to get clever like with Example 3.7 or Example 4.3.
- Any time a concurrency and collinearity problem has one or more important circles, you should think of radical axes/center. If the problem statement asks you to show that three lines are "all parallel or concurrent," that's also a big red flag.
- Point circles may occasionally be useful for problems involving lots of equal lengths, especially if there are perpendicular bisectors.
- Power of a Point (and especially linearity) can be helpful for problems involving fixed points or circles as we saw in Example 5.8; showing that the power of some point is fixed is a pretty underrated tool in my opinion.

§6.2 Further Reading

For more reading on length chasing, see AoPS user mira74's Ratio Lemma handout or SinaQane's length bashing formula sheet. The topic of geometric inversion is also closely related to Power of a Point, for which Gunmay Handa's inversion handout (download link!) and Chapter 8 of Evan Chen's textbook *Euclidean Geometry in Mathematical Olympiads* are useful references. Another useful article on an underrated geometric transformation is Radek Olšák's polar duality handout.

§7 Problems

Problems in each section are roughly sorted by difficulty. Please only refer to the hints if you are well and truly stuck, as many are giveaways! I've also linked AoPS threads for most of the problems, and once you solve or simply cannot make any more progress on a problem, feel free to check out other people's solutions; I personally have also posted solutions on many of the threads.

§7.1 Introduction

Problem 7.1. Let AD be the A-altitude in triangle $\triangle ABC$. If H is a point on AD, prove that H is the orthocenter of $\triangle ABC$ if and only if $BD \cdot CD = AD \cdot AH$.

Problem 7.2 (AoPS)**.** Let BT be the altitude and H be the intersection point of the altitudes of triangle ABC. Point N is symmetric to H with respect to BC. The circumcircle of triangle ATN intersects BC at points F and K. Prove that $FB = BK$.

Problem 7.3 (Sharygin CR 2021/16)**.** Let circles Ω and ω touch internally at point A. A chord BC of Ω touches ω at point K. Let O be the center of ω. Prove that the circle BOC bisects segment AK.

Problem 7.4 (USAMO 1998/2)**.** Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be concentric circles, with $\mathcal{C}_2$ in the interior of $\mathcal{C}_1$. From a point A on $\mathcal{C}_1$ one draws the tangent AB to $\mathcal{C}_2$ ($B \in \mathcal{C}_2$). Let C be the second point of intersection of ray AB and $\mathcal{C}_1$, and let D be the midpoint of AB. A line passing through A intersects $\mathcal{C}_2$ at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB. Find, with proof, the ratio $\frac{AM}{MC}$.

Problem 7.5 (Canadian MO 2013/3)**.** Let G be the centroid of a right-angled triangle ABC with $\angle BCA = 90^\circ$. Let P be the point on ray AG such that $\angle CPA = \angle CAB$, and let Q be the point on ray BG such that $\angle CQB = \angle ABC$. Prove that the circumcircles of triangles AQG and BPG meet at a point on side AB.

Problem 7.6 (AIME I 2020/15)**.** Let ABC be an acute triangle with circumcircle ω and orthocenter H. Suppose the tangent to the circumcircle of $\triangle HBC$ at H intersects ω at points X and Y with $HA = 3$, $HX = 2$, $HY = 6$. The area of $\triangle ABC$ can be written as $m\sqrt{n}$, where m and n are positive integers, and n is not divisible by the square of any prime. Find $m + n$.

Problem 7.7 (IMO 2019/2)**.** In triangle ABC, point A_1 lies on side BC and point B_1 lies on side AC. Let P and Q be points on segments AA_1 and BB_1, respectively, such that PQ is parallel to AB. Let P_1 be a point on line PB_1, such that B_1 lies strictly between P and P_1, and $\angle PP_1C = \angle BAC$. Similarly, let Q_1 be the point on line QA_1, such that A_1 lies strictly between Q and Q_1, and $\angle CQ_1Q = \angle CBA$. Prove that points P, Q, P_1, and Q_1 are concyclic.

Problem 7.8 (IMO Shortlist 2008/G4)**.** In an acute triangle ABC segments BE and CF are altitudes. Two circles passing through the point A and F and tangent to the line BC at the points P and Q so that B lies between C and Q. Prove that lines PE and QF intersect on the circumcircle of triangle AEF.

Problem 7.9 (IMO 2009/2)**.** Let ABC be a triangle with circumcentre O. The points P and Q are interior points of the sides CA and AB respectively. Let K, L and M be the midpoints of the segments BP, CQ, and PQ. respectively, and let Γ be the circle passing through K, L and M. Suppose that the line PQ is tangent to the circle Γ. Prove that $OP = OQ$.

§7.2 Radical Axis

Problem 7.10 (Orthocenter)**.** Prove that the three altitudes of a triangle are concurrent.

Problem 7.11 (APMO 2020/1)**.** Let Γ be the circumcircle of $\triangle ABC$. Let D be a point on the side BC. The tangent to Γ at A intersects the parallel line to BA through D at point E. The segment CE intersects Γ again at F. Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.

Problem 7.12 (AoPS)**.** Let A, B, C, D be points in the plane such that CD is a line perpendicular to AB and $C, D \notin AB$. If E and F are the orthocenters of $\triangle ABC$ and $\triangle ABD$, prove that the radical axis of the circles with diameters CE and DF is AB.

Problem 7.13 (AoPS)**.** In $\triangle ABC$, let I be the incenter and D, E, F be the touch points of the incircle with BC, CA, AB, respectively. If AI meets (ABC) again at M and K, Q are the points where the tangent to (ABC) at M meets AC, AB, respectively, show that D lies on the radical axis of (EIK) and (FIQ).

Problem 7.14 (Orthic Axis)**.** Let AD, BE, CF be altitudes in $\triangle ABC$, and let $A_1 = BC \cap EF, B_1 = CA \cap FD$, and $C_1 = AB \cap DE$. Prove that A_1, B_1, C_1 lie on a line perpendicular to the Euler Line of $\triangle ABC$.

Problem 7.15 (IMO 2008/1)**.** Let H be the orthocenter of an acute-angled triangle ABC. The circle Γ_A centered at the midpoint of BC and passing through H intersects the sideline BC at points A_1 and A_2. Similarly, define the points B_1, B_2, C_1, and C_2. Prove that six points A_1, A_2, B_1, B_2, C_1, and C_2 are concyclic.

Problem 7.16 (PUMaC Finals 2017/A3)**.** Triangle ABC has incenter I. The line through I perpendicular to AI meets the circumcircle of ABC at points P and Q, where P and B are on the same side of AI. Let X be the point such that $PX||CI$ and $QX||BI$. Show that PB, QC, and IX intersect at a common point.

Problem 7.17 (Sharygin CR 2018/20, GOTEEM 2020/1)**.** Let the incircle of a non-isosceles triangle ABC touch AB, AC and BC at points D, E and F respectively. The corresponding excircle touches the side BC at point N. Let T be the common point of AN and the incircle, closest to N, and K be the common point of DE and FT. Prove that $AK||BC$.

Problem 7.18 (TSTST 2017/1)**.** Let ABC be a triangle with circumcircle Γ, circumcenter O, and orthocenter H. Assume that $AB \neq AC$ and that $\angle A \neq 90°$. Let M and N be the midpoints of sides AB and AC, respectively, and let E and F be the feet of the altitudes from B and C in $\triangle ABC$, respectively. Let P be the intersection of line MN with the tangent line to Γ at A. Let Q be the intersection point, other than A, of Γ with the circumcircle of $\triangle AEF$. Let R be the intersection of lines AQ and EF. Prove that $PR \perp OH$.

Problem 7.19 (EGMO 2014/2)**.** Let D and E be points in the interiors of sides AB and AC, respectively, of a triangle ABC, such that $DB = BC = CE$. Let the lines CD and BE meet at F. Prove that the incentre I of triangle ABC, the orthocentre H of triangle DEF and the midpoint M of the arc BAC of the circumcircle of triangle ABC are collinear.

Problem 7.20 (IMO 2012/5)**.** Let ABC be a triangle with $\angle BCA = 90°$, and let D be the foot of the altitude from C. Let X be a point in the interior of the segment CD. Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK. Show that $MK = ML$.

§7.3 Coaxial Circles

Problem 7.21. Let AD, BE, CF be the altitudes of a scalene triangle with circumcenter O. Prove that (AOD), (BOE), and (COF) are coaxial and furthermore that their common radical axis is the Euler Line.

Problem 7.22 (AoPS)**.** In scalene $\triangle ABC$ with circumcircle Γ, let D be the midpoint of BC. Line $\overline{AD}$ meets Γ again at A_1. The tangent to Γ at A_1 intersects $\overline{BC}$ at A_2. Denote by ω_A the circumcircle of $\triangle DA_1A_2$, and define ω_B, ω_C similarly. Prove that ω_A, ω_B, and ω_C are coaxial.

Problem 7.23 (AoPS)**.** In $\triangle ABC$, let D, E, and F be the touch points of the incircle with BC, CA, and AB, respectively. If P is the foot from D to EF and $X = AB \cap CP$ and $Y = AC \cap BP$, then prove that $(AXY), (AEF)$, and (ABC) meet at a point other than A.

Problem 7.24 (IMO Shortlist 2005/G5)**.** Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC, and let M be the midpoint of the side BC. Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.

Problem 7.25 (SMO 2020/5)**.** In triangle $\triangle ABC$, let E and F be points on sides AC and AB, respectively, such that $BFEC$ is cyclic. Let lines BE and CF intersect at point P, and M and N be the midpoints of $\overline{BF}$ and $\overline{CE}$, respectively. If U is the foot of the perpendicular from P to BC, and the circumcircles of triangles $\triangle BMU$ and $\triangle CNU$ intersect at second point V different from U, prove that A, P, and V are collinear.

Problem 7.26 (European Mathematical Cup 2016/4)**.** Let C_1, C_2 be circles intersecting in X, Y . Let A, D be points on C_1 and B, C on C_2 such that A, X, C are collinear and D, X, B are collinear. The tangent to circle C_1 at D intersects BC and the tangent to C_2 at B in P, R respectively. The tangent to C_2 at C intersects AD and tangent to C_1 at A, in Q, S respectively. Let W be the intersection of AD with the tangent to C_2 at B and Z the intersection of BC with the tangent to C_1 at A. Prove that the circumcircles of triangles YWZ, RSY and PQY have two points in common, or are tangent in the same point.

Problem 7.27 (IMO 1979/3)**.** Two circles in a plane intersect. A is one of the points of intersection. Starting simultaneously from A two points move with constant speed, each travelling along its own circle in the same sense. The two points return to A simultaneously after one revolution. Prove that there is a fixed point P in the plane such that the two points are always equidistant from P.

Problem 7.28 (AoPS, taken from here)**.** Let ABC be a triangle and let L, M, N be points on sides BC, CA, AB respectively. Let Q be the Miquel point of L, M, N with respect to $\triangle ABC$. Show that if AL, BM, CN concur, then (AQL), (BQM), (CQN) are coaxial.

§7.4 Point Circles

Problem 7.29 (Iran TST 2011/1)**.** In acute triangle ABC, $\angle B$ is greater than $\angle C$. Let M be the midpoint of BC and let E and F be the feet of the altitudes from B and C, respectively. Let K and L be the midpoints of ME and MF, respectively, and let T be on line KL such that $TA \parallel BC$. Prove that $TA = TM$.

Problem 7.30 (Iran MO 2017 Round 3/G3)**.** Let ABC be an acute-angle triangle. Suppose that M be the midpoint of BC and H be the orthocenter of ABC. Let $F \equiv BH \cap AC$ and $E \equiv CH \cap AB$. Suppose that X be a point on EF such that $\angle XMH = \angle HAM$ and A, X are in the distinct side of MH. Prove that AH bisects MX.

Problem 7.31 (2015 Sharygin CR/19)**.** Let L and K be the feet of the internal and the external bisector of angle A of a triangle ABC. Let P be the common point of the tangents to the circumcircle of the triangle at B and C. The perpendicular from L to BC meets AP at point Q. Prove that Q lies on the medial line of triangle LKP.

Problem 7.32 (IMO Shortlist 2009/G3)**.** Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y, respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelogram. Prove that $GR = GS$.

Problem 7.33 (ARMO 2011 Grade 10/4)**.** The perimeter of triangle ABC is 4. Point X is marked on ray AB and point Y is marked on ray AC such that $AX = AY = 1$. If BC intersects XY at point M, prove that perimeter of one of triangles ABM or ACM is 2.

Problem 7.34 (NICE MO 2021/2)**.** Let O be the circumcenter of triangle ABC. Suppose the perpendicular bisectors of $\overline{OB}$ and $\overline{OC}$ intersect lines AB and AC at $D \neq A$ and $E \neq A$, respectively. Determine the maximum possible number of distinct intersection points between line BC and the circumcircle of $\triangle ADE$.

§7.5 Black Magic

Problem 7.35 (IMO Shortlist 2011/G2)**.** Let $A_1A_2A_3A_4$ be a non-cyclic quadrilateral. Let O_1 and r_1 be the circumcentre and the circumradius of the triangle $A_2A_3A_4$. Define O_2, O_3, O_4 and r_2, r_3, r_4 in a similar way. Prove that

$$\frac{1}{O_1A_1^2 - r_1^2} + \frac{1}{O_2A_2^2 - r_2^2} + \frac{1}{O_3A_3^2 - r_3^2} + \frac{1}{O_4A_4^2 - r_4^2} = 0.$$

Problem 7.36. In triangle $\triangle ABC$, let O, G, and K be the circumcenter, centroid, and Symmedian Point, respectively. Prove that $OK \geq OG$.

Problem 7.37 (Taiwan TST 2016 Round 2/1)**.** Let O be the circumcenter of triangle ABC, and ω be the circumcircle of triangle BOC. Line AO intersects with circle ω again at the point G. Let M be the midpoint of side BC, and the perpendicular bisector of BC meets circle ω at the points O and N. Prove that the midpoint of the segment AN lies on the radical axis of the circumcircle of triangle OMG, and the circle whose diameter is AO.

Problem 7.38 (ELMO Shortlist 2013/G3)**.** In $\triangle ABC$, a point D lies on line BC. The circumcircle of ABD meets AC at F (other than A), and the circumcircle of ADC meets AB at E (other than A). Prove that as D varies, the circumcircle of AEF always passes through a fixed point other than A, and that this point lies on the median from A to BC.

Problem 7.39 (USA December TST 2012/1)**.** In acute triangle ABC, $\angle A < \angle B$ and $\angle A < \angle C$. Let P be a variable point on side BC. Points D and E lie on sides AB and AC, respectively, such that $BP = PD$ and $CP = PE$. Prove that as P moves along side BC, the circumcircle of triangle ADE passes through a fixed point other than A.

§7.6 Challenge Problems

Problem 7.40 (Haruki's Lemma)**.** Let AB, CD be two fixed chords and P be a variable point on a circle. If $Q = PC \cap AB$ and $R = PD \cap AB$, prove that the quantity $\frac{AQ \cdot RB}{QR}$ does not depend on P. As a bonus, try using this lemma to prove the Butterfly Theorem!

Problem 7.41 (USA TST 2006/6)**.** Let ABC be a triangle. Triangles PAB and QAC are constructed outside of triangle ABC such that $AP = AB$ and $AQ = AC$ and $\angle BAP = \angle CAQ$. Segments BQ and CP meet at R. Let O be the circumcenter of triangle BCR. Prove that $AO \perp PQ$.

Problem 7.42 (Iran MO 2003 Round 3/P6)**.** Let the incircle of a triangle ABC touch BC, AC, AB at A_1, B_1, C_1 respectively. Let M and N be the midpoints of AB_1 and AC_1, respectively, and MN meet A_1C_1 at T. Draw two tangents TP and TQ through T to incircle. PQ meets MN at L and B_1C_1 meets PQ at K. If I is the center of the incircle, prove that IK is parallel to AL.

Problem 7.43 (USA December TST 2020/2)**.** Two circles Γ_1 and Γ_2 have common external tangents ℓ_1 and ℓ_2 meeting at T. Suppose ℓ_1 touches Γ_1 at A and ℓ_2 touches Γ_2 at B. A circle Ω through A and B intersects Γ_1 again at C and Γ_2 again at D, such that quadrilateral $ABCD$ is convex. Suppose lines AC and BD meet at point X, while lines AD and BC meet at point Y. Show that T, X, Y are collinear.

Problem 7.44 (Crucial lemma of Sharygin CR 2021/18)**.** Let $\triangle ABC$ be a triangle with incenter I, let D be some point on BC, let K be the midpoint of ID, and $P = AI \cap (ABC) \neq A$. If ω_1 is the circle tangent to segments AD, BD, and circle (ABC) internally and ω_2 is the circle tangent to segments AD, CD, and (ABC) internally, prove that PK is the radical axis of ω_1 and ω_2.

Problem 7.45 (USA February TST 2021/2)**.** Points A, V_1, V_2, B, U_2, U_1 lie fixed on a circle Γ, in that order, and such that $BU_2 > AU_1 > BV_2 > AV_1$. Let X be a variable point on the arc V_1V_2 of Γ not containing A or B. Line XA meets line U_1V_1 at C, while line XB meets line U_2V_2 at D. Let O and ρ denote the circumcenter and circumradius of $\triangle XCD$, respectively. Prove there exists a fixed point K and a real number c, independent of X, for which $OK^2 - \rho^2 = c$ always holds regardless of the choice of X.

Problem 7.46 (IMO Shortlist 2017/G7)**.** A convex quadrilateral $ABCD$ has an inscribed circle with center I. Let I_a, I_b, I_c and I_d be the incenters of the triangles DAB, ABC, BCD and CDA, respectively. Suppose that the common external tangents of the circles AI_bI_d and CI_bI_d meet at X, and the common external tangents of the circles BI_aI_c and DI_aI_c meet at Y. Prove that $\angle XIY = 90°$.

Problem 7.47 (USEMO 2019/6)**.** Let ABC be an acute scalene triangle with circumcenter O and altitudes $\overline{AD}$, $\overline{BE}$, $\overline{CF}$. Let X, Y, Z be the midpoints of $\overline{AD}$, $\overline{BE}$, $\overline{CF}$. Lines AD and YZ intersect at P, lines BE and ZX intersect at Q, and lines CF and XY intersect at R. Suppose that lines YZ and BC intersect at A', and lines QR and EF intersect at D'. Prove that the perpendiculars from A, B, C, O, to the lines QR, RP, PQ, $A'D'$, respectively, are concurrent.

Problem 7.48 (IMO 2019/6)**.** Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA, and AB at D, E, and F, respectively. The line through D perpendicular to EF meets ω at R. Line AR meets ω again at P. The circumcircles of triangle PCE and PBF meet again at Q. Prove that lines DI and PQ meet on the line through A perpendicular to AI.

§8 Hints

7.1. Reflect H over BC.

7.2. Reflect H over B.

7.3. Draw the intersection of BC and the common tangent at A.

7.4. Proving that $DEFC$ is cyclic implies that M is the midpoint of CD.

7.5. Show that (CGP) is tangent to AC.

7.6. Reflect H over BC and construct the tangent to ω at that point.

7.7. Extend AA_1 and BB_1 to hit the circumcircle again.

7.8. Prove that $AEPQ$ is cyclic, then angle chase.

7.9. Show that $\triangle ABQ \stackrel{-}{\sim} \triangle MKL$.

7.10. H is the radical center of three circles.

7.12. Let $P = AB \cap CD$, and use PoP.

7.13. KI is the diameter of (EIK).

7.14. Show that A_1, B_1, C_1 lie on the radical axis of the nine-point circle and (ABC).

7.15. Either show that if $(A_1A_2C_1C_2) \neq (B_1B_2C_1C_2)$, then C would lie on their radical axis, or show that $(A_1A_2C_1C_2), (B_1B_2C_1C_2), (C_1C_2A_1A_2)$ share a common center.

7.16. Draw $D = PX \cap CQ, E = QX \cap BP$. Then use radical center twice.

7.17. Radical center on $(AEF), (DEF), (ADT)$.

7.18. Use radical center twice to show that P and R lie on the radical axis of (ABC) and the nine-point circle.

7.19. M, I, H form the radical axis of two circles.

7.20. Introduce E and F, the second intersections of BX with the circle centered at A with radius AC and AX with the circle centered at B with radius BC, respectively.

7.21. Show that H lies on the pairwise radical axes.

7.22. Prove that the circumcenter and centroid lie on the radical axes.

7.23. Use the Forgotten Coaxiality Lemma on X, Y with respect to $(AEF), (ABC)$.

7.24. Use the Forgotten Coaxiality lemma on D, E with respect to $(AH), (ABC)$.

7.25. Use the Forgotten Coaxiality Lemma on $P, Q = EF \cap BC$ with respect to $(BMU), (CMU)$.

7.26. Use the Forgotten Coaxiality Lemma on W, Z with respect to (PQY) and (RSY).

7.27. If the two points are X and Y, prove that the midpoint of XY lies on a fixed circle with the Forgotten Coaxiality Lemma.

7.28. Use the Forgotten Coaxiality Lemma 5 times.

7.29. Consider the radical axis of M and (AEF).

7.30. $AH \cap MX$ is the radical center of M and two other circles.

7.31. Q is the radical center of L and two other circles.

7.32. Introduce the A-excircle; there are lots of equal lengths that result.

7.33. Reflect X over B and Y over C to get the tangency points of the A-excircle.

7.34. Consider the radical axis of O and (ADE).

7.35. Pick any 3 of the A_i as your reference triangle, then just use the bary PoP formula.

7.36. It suffices to prove that $\text{Pow}(O, (ABC)) \geq \text{Pow}(G, (ABC))$, which can be done with either straight bary or Example 5.3.

7.37. Plug P into $f(\bullet) = \text{Pow}(\bullet, (OMG)) - \text{Pow}(\bullet, (AO))$.

7.38. Show that $\text{Pow}(M, (AEF))$ is fixed.

7.39. Show that $\text{Pow}(H, (ADE))$ is fixed using $H = (\tan A : \tan B : \tan C)$.

7.40. Extend AB to meet (PQD) at E. Is there anything special about the length of BE?

7.41. Introduce $X = AP \cap (ACQR), Y = AP \cap (ABPR)$. Prove that $PO^2 - PA^2 = \text{Pow}(A, (PQX))$.

7.42. L is the radical center of $A, (APE)$, and the incircle.

7.43. Consider the radical axis of (ADE) and (FCB).

7.44. For P, simple PoP manipulations do the trick. For K, let I_1, I_2 be the centers of ω_1, ω_2, and let V be the reflection of D over M. If $f(\bullet) = \text{Pow}(\bullet, \omega_1) - \text{Pow}(\bullet, \omega_2)$, after some manipulations of the equation $f(K) = \frac{1}{2}(f(I) + f(D)) = 0$, it suffices to show that $IV \perp I_1I_2$. Also, don't forget the Sawayama-Thebault Theorem!

7.45. Let $A' = A\infty_{U_1V_1} \cap \Gamma \neq A$ and $B' = B\infty_{U_2V_2} \cap \Gamma \neq B$. Then prove that $K = AB' \cap AA'$ is fixed with the function $\text{Pow}(\bullet, (XCD)) - \text{Pow}(\bullet, \Gamma)$.

7.46. Use the Forgotten Coaxiality Lemma on X, I with respect to $(AI_bI_c), (CI_bI_c)$, and use XII_bI_c cyclic to show that XI bisects $\angle I_bII_c$; symmetry finishes the problem from there.

7.47. Prove that $A'D'$ is the radical axis of (DEF) and (XYZ).

7.48. Define $f_1(\bullet) = \text{Pow}(\bullet, (BDIF)) - \text{Pow}(\bullet, (CDIE))$ and $f_2(\bullet) = \text{Pow}(\bullet, (BPF)) - \text{Pow}(\bullet, (CPE))$. The crucial step is proving that for any point T on AX, $f_1(T) = f_2(T)$.

§9 Appendices

§9.1 Appendix A: Notations and conventions

- $\cap$ denotes the intersection of two objects. For example, if a point P is defined as the intersection of two lines ℓ_1 and ℓ_2, it would be written as $P = \ell_1 \cap \ell_2$.
- $\in$ denotes that a point is on a figure. For example, if a point P lies on circle ω, it would be written as $P \in \omega$.
- Parentheses denote the circumcircle of a figure or the diameter circle if we only list two points. For example, the circumcircle of a triangle $\triangle ABC$ is (ABC), and the circle with diameter AB is (AB).
- The abbreviation PoP means "Power of a Point," and WLOG means "Without Loss Of Generality." I have no idea why the O is inconsistently capitalized either.
- The circumradius of the reference triangle (usually $\triangle ABC$) is denoted by R, and the inradius is denoted by r. Also, in $\triangle ABC$, a, b, and c denote the lengths of sides BC, CA, and AB, respectively, and we use $\angle A, \angle B$, and $\angle C$ as shorthand for $\angle CAB, \angle ABC$, and $\angle BCA$, respectively.
- The symbol $\overset{+}{\sim}$ means directly similar, and the symbol $\overset{-}{\sim}$ means inversely similar. For example, if $\triangle ABC \overset{+}{\sim} \triangle DEF$, it means that in terms of directed angles, $\measuredangle ABC = \measuredangle DEF$, while if $\triangle ABC \overset{-}{\sim} \triangle DEF$, it means that $\measuredangle ABC = \measuredangle FED$.
- ∞_ℓ denotes the point at infinity on a line ℓ, i.e. the point that all lines parallel to ℓ pass through. In particular, $P\infty_\ell$ is the line through point P parallel to line ℓ.

§9.2 Appendix B: Referenced theorems without proof

Theorem 9.1 (Angle Bisector Theorem)

In triangle $\triangle ABC$, let D be a point on segment BC. Then $\frac{AB}{AC} = \frac{BD}{CD}$ if and only if D lies on the A-internal angle bisector. If D is outside of segment BC, then $\frac{AB}{AC} = \frac{BD}{CD}$ if and only if D lies on the A-external angle bisector.

Theorem 9.2 (Reim's Theorem)

Let $ABPQ$ be a cyclic quadrilateral. If C and D are two points on lines AQ and BP, respectively, then $ABCD$ is cyclic if and only if $CD \parallel PQ$.

Theorem 9.3 (Menelaus's Theorem)

In triangle $\triangle ABC$, let D, E, F be points on lines BC, CA, AB, respectively. Then D, E, F are collinear if and only if

$$\frac{BD}{CD} \cdot \frac{CE}{AE} \cdot \frac{AF}{BF} = -1,$$

where lengths are directed (i.e. $\frac{DB}{DC}$ is positive if $\vec{DB}$ is in the same direction as $\vec{DC}$ and negative otherwise).

Theorem 9.4 (Stewart's Theorem)
In triangle $\triangle ABC$, let D be a point on BC. Then if $BD = m$ and $CD = n$, we have

$$a(mn + d^2) = b^2m + c^2n \iff man + dad = bmb + cnc.$$

Theorem 9.5 (Law of Sines)
In triangle $\triangle ABC$, we have

$$\frac{a}{\sin \angle A} = \frac{b}{\sin \angle B} = \frac{c}{\sin \angle C} = 2R.$$

Theorem 9.6 (Law of Cosines)
In triangle $\triangle ABC$, we have

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Theorem 9.7 (Miquel Pivot Theorem)
In triangle $\triangle ABC$, let D, E, F be points on BC, CA, AB, respectively. Then circles $(AEF), (BFD), (CDE)$ have one point in common, the **Miquel Point** of D, E, F with respect to $\triangle ABC$.

Theorem 9.8 (Schwatt Line)
In triangle $\triangle ABC$, let X_A, X_B, X_C be the midpoints of the A-, B-, C-altitudes, respectively, and let M_A, M_B, M_C be the midpoints of BC, CA, AB, respectively. Then X_AM_A, X_BM_B, X_CM_C concur at the Symmedian Point of $\triangle ABC$.

§9.3 Appendix C: Trigonometric bash for TSTST 2016/6

Claim 9.9 — We wish to show that

$$\frac{CD^2}{BD^2} \cdot \frac{\operatorname{dist}(G, D_0E_0)}{\operatorname{dist}(G, F_0D_0)} \cdot \frac{\operatorname{dist}(B, F_0D_0)}{\operatorname{dist}(C, D_0E_0)} = 1.$$

Proof. Abbreviate $\angle D = \angle EDF, \angle E = \angle FED$, and $\angle F = \angle DFE$. For the first quotient, we have

$$\frac{CD^2}{BD^2} = \left(\frac{\frac{1}{2}DE}{\frac{1}{2}FD} \cdot \frac{\cos(90^\circ - \frac{1}{2}\angle CBA)}{\cos(90^\circ - \frac{1}{2}\angle ACB)}\right)^2 = \frac{DE^2}{FD^2} \cdot \frac{\cos^2 \angle E}{\cos^2 \angle F}.$$

For the second quotient, it's well-known that the Gergonne Point is the isotomic conjugate of the orthocenter in $\triangle D_0E_0F_0$, so the barycentric coordinates of G in $\triangle D_0E_0F_0$ are $(\cot \angle D : \cot \angle E : \cot \angle F)$, giving

$$\frac{\operatorname{dist}(G, D_0E_0)}{\operatorname{dist}(G, F_0D_0)} = \frac{[GD_0E_0]}{[GF_0D_0]} \cdot \frac{\frac{1}{D_0E_0}}{\frac{1}{F_0D_0}} = \frac{\cot \angle F}{\cot \angle E} \cdot \frac{\sin \angle E}{\sin \angle F} = \frac{\cos \angle F}{\cos \angle E} \cdot \frac{FD^2}{DE^2}$$

with the second and third equalities coming from the Law of Sines on $\triangle D_0E_0F_0$ and $\triangle DEF$, respectively.

For the third quotient, we have

$$\begin{aligned}
\operatorname{dist}(B, F_0D_0) &= BD\sin\angle E + \frac{1}{2}DE\sin\angle D \\
&= BD\sin\angle E + \frac{DE\cdot EF}{2FD}\sin\angle E \\
&= \sin\angle E\left(\frac{FD}{2\cos\angle E} + \frac{DE\cdot EF}{2FD}\right) \\
&= \frac{\sin\angle E}{2FD\cos\angle E}\left(FD^2 + DE\cdot EF\cdot\cos\angle E\right) \\
&= \frac{\sin\angle E}{2FD\cos\angle E}\left(FD^2 + DE\cdot EF\cdot\frac{DE^2+EF^2-FD^2}{2DE\cdot EF}\right) \\
&= \frac{\sin\angle E}{4FD\cos\angle E}\left(FD^2 + DE^2 + EF^2\right),
\end{aligned}$$

where the second and fifth equality come from Law of Sines and Law of Cosines, respectively, on $\triangle DEF$. Thus,

$$\frac{\operatorname{dist}(B, F_0D_0)}{\operatorname{dist}(C, D_0E_0)} = \frac{\cos\angle F}{\cos\angle E}\cdot\frac{FD\sin\angle E}{DE\sin\angle F} = \frac{\cos\angle F}{\cos\angle E},$$

with the second equality coming from Law of Sines on $\triangle DEF$, so multiplying all three quotients yields 1 as desired. $\square$